

Curves and maps — From families to stacks

Thesis study notes

Ignacio Rojas

ChatGPT (OpenAI)

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Definitions, descent, and the geometric input behind the constructions

Developed from Ignacio Rojas's thesis questions, with exposition, source checking, and drafting by ChatGPT (OpenAI). The purpose is to make each categorical and gluing step usable before compressing it into the statement that a moduli problem is a stack.

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Chapter 1

$\overline{\mathcal{M}}_{g,n}$ is a stack

1.1 A family of stable pointed curves

Start with a scheme S over $\mathbb{C}$, and fix integers $g, n \geq 0$ with

$$2g - 2 + n > 0. \tag{1.1.1}$$

We want to put one complete, connected, possibly nodal curve over each point of S , together with n labelled points that vary as sections. The scheme S is the parameter scheme; it is not the curve being classified. The inequality is the *stable range*: for example, $(g, n) = (2, 0), (1, 1), (0, 3)$ are allowed, and $(0, 0)$ is not allowed for stable pointed curves.

Definition 1.1 (Family of stable pointed curves). The input is a $\mathbb{C}$ -scheme S . A family over S consists of a scheme C , a proper flat finitely presented morphism $\pi : C \rightarrow S$, and sections $\sigma_a : S \rightarrow C$ for $1 \leq a \leq n$, such that:

- (a) For every geometric point $\bar{s} : \text{Spec } \Omega \rightarrow S$, where Ω is algebraically closed, the fiber $C_{\bar{s}} = C \times_S \text{Spec } \Omega$ is a connected reduced curve with only ordinary nodes as possible singularities, and arithmetic genus g .
- (b) Each section takes values in the smooth locus of π , and the images of different sections are disjoint.
- (c) In the normalization of each geometric fiber, every component of genus zero carries at least three special points, and every component of genus one carries at least one special point. A special point is a marked point or a preimage of a node; a self-node contributes two such preimages to the same component.

We write $\xi = (S, C, \pi, \sigma_1, \dots, \sigma_n)$ for all these data.

Here “finitely presented” rules out infinitely many equations or parameters in the relative construction; it is the useful finiteness condition when S need not be noetherian. Flatness is a condition on the family, not just on its individual fibers. Properness is the algebraic completeness condition. The sections satisfy $\pi\sigma_a = \text{id}_S$ and remember their labels. Their disjointness is a condition on the entire family, including nonreduced bases.

At a node of a geometric fiber, the completed local ring is $\Omega[[x, y]]/(xy)$: two smooth branches meet. The arithmetic genus of a proper connected reduced curve over Ω is $g = \dim_{\Omega} H^1(C_{\bar{s}}, \mathcal{O}_{C_{\bar{s}}})$. Thus a genus-two fiber need not be a smooth surface with two handles; it can be reducible, with the genus distributed among its components and its incidence graph.

The normalization separates the branches at each node. If a normalized component E has genus h and r special points, the stability test can be written as $2h - 2 + r > 0$. In particular, its content concerns *each component*, not just the total genus. The geometric facts relating this test to automorphisms and line bundles are explained in Alper [4, Definition 6.3.2, Proposition 6.3.5, pp. 225–226]. The family definition is [4, Definition 6.3.20, p. 231].

Example 1.1 (A smooth genus-two object). Let C_0 be the smooth projective completion of

$$y^2 = x^6 - 1$$

over $\mathbb{C}$. More precisely it is the smooth double cover of $\mathbb{P}^1$ branched at the six distinct sixth roots of unity. The double-cover genus formula gives $2g(C_0) - 2 = 2(-2) + 6 = 2$, hence $g(C_0) = 2$. The affine equation alone would not be a proper curve, which is why the completion is part of the description.

With no markings, $C_0 \rightarrow \text{Spec } \mathbb{C}$ is an object for $(g, n) = (2, 0)$. Its only normalized component has $h = 2, r = 0$, so it is stable. The involution $\iota(x, y) = (x, -y)$ extends over the completion. Stability allows this nontrivial automorphism; it asks for a finite automorphism group, not a trivial group. If we want one marking, $p = (1, 0)$ is a smooth point fixed by ι , so (C_0, p) is a useful example for $(2, 1)$ too.

Example 1.2 (A stable nodal genus-two object). Take smooth elliptic curves E_1, E_2 with chosen points q_1, q_2 , and identify these points to one ordinary node q . The result $C = E_1 \cup_q E_2$ is projective and connected. The normalization sequence is

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{O}_{E_1} \oplus \mathcal{O}_{E_2} \longrightarrow \mathbb{C}_q \longrightarrow 0,$$

where the last map takes the difference of values at the two branches. It gives $\chi(\mathcal{O}_C) = 0 + 0 - 1 = -1$, hence genus two. Each normalized elliptic component has its one node preimage as a special point. Thus both have $2h - 2 + r = 1 > 0$. This is a stable unmarked curve even though neither elliptic component would be stable as an isolated unmarked curve.

Exercise 1.1. A rational component has one self-node and one additional marked smooth point. How many special points are on its normalization? Does this component pass the stability test? Why would counting the self-node just once give the wrong answer?

1.1.1 Base and notation for the whole construction

Our base category is $\text{Sch}/\mathbb{C}$, the category of all $\mathbb{C}$ -schemes and $\mathbb{C}$ -morphisms, with the *étale topology*. A cover is a family of étale maps $\{S_i \rightarrow S\}$ whose images together cover S . Ordinary open covers are examples, but a cover need not consist of inclusions. For example,

$$\text{Spec } \mathbb{C}[u, u^{-1}] \longrightarrow \text{Spec } \mathbb{C}[t, t^{-1}], \quad t = u^2,$$

is a one-map étale cover. Locally it supplies a choice of square root; globally there is no preferred choice.

We use $\overline{\mathcal{M}}_{g,n}$ for the stable pointed-curve construction throughout. The notation $\mathcal{M}_{g,n}$ denotes its open substack of *smooth* curves, once the stack has been constructed. With $n = 0$, write $\overline{\mathcal{M}}_g$ and $\mathcal{M}_g$, respectively. An informal “ M_g ” must be disambiguated: smooth curves and their stable compactification are different moduli problems. Plain noncalligraphic symbols in some references instead denote coarse spaces; we do not use them for our stacks.

1.2 First construct an ordinary category

1.2.1 Objects and arrows

The objects of our *total category* are all the tuples ξ of Definition 1.1, allowing S to vary. Thus the base is part of the object. The same curve family pulled back to a new base is a new object, even if its fibers look familiar.

Definition 1.2 (An arrow between families). Given objects $\xi = (C \xrightarrow{\pi} S, \sigma_\bullet)$ and $\eta = (D \xrightarrow{\rho} T, \tau_\bullet)$, an arrow $(u, h) : \xi \rightarrow \eta$ consists of maps $u : S \rightarrow T$ and $h : C \rightarrow D$

such that

$$\begin{array}{ccc}
 C & \xrightarrow{h} & D \\
 \pi \downarrow & & \downarrow \rho \\
 S & \xrightarrow{u} & T
 \end{array}
 \quad \rho h = u\pi, \quad h\sigma_a = \tau_a u \quad (1 \leq a \leq n),$$

and the square is cartesian. Explicitly, the induced map

$$(\pi, h) : C \longrightarrow S \times_T D$$

must be an isomorphism.

The input is two families. The data of an arrow say that the first is the second viewed over the new base S , with an identification of the pulled-back total space. The condition is an isomorphism of *schemes over S* , not merely a bijection of geometric fibers. The map $h : C \rightarrow D$ itself need not be an isomorphism when u is not one. For instance, inclusion of a fiber is usually not an isomorphism of total spaces.

Why not allow arbitrary maps between curves? The moduli problem identifies two presentations of the same family. A degree-two map between curves or a constant map changes the geometric object; it is not such an identification. In particular, a constant $C_0 \rightarrow C_0$ is not an arrow here. This choice of arrows is essential to obtaining groupoids as fibers. It does not mean that every arrow of the total category is invertible.

Example 1.3 (Arrows over a point). Over $\text{Spec } \mathbb{C}$, the only base endomorphism as a $\mathbb{C}$ -scheme is the identity. An arrow from C_0 to a second genus-two curve D_0 is therefore exactly a curve isomorphism $h : C_0 \xrightarrow{\sim} D_0$. If points are marked, h must send each labelled point to the point with the same label. The identity and ι give different arrows $C_0 \rightarrow C_0$, including in the one-pointed example $(C_0, (1, 0))$. An object and its automorphisms are retained, not replaced by just one element of a set of isomorphism classes.

1.2.2 Identities, composition, and the axioms

The identity of ξ is $(\text{id}_S, \text{id}_C)$. Its square is cartesian and it preserves the sections. If $(u, h) : \xi \rightarrow \eta$ and $(v, k) : \eta \rightarrow \zeta$, where $\zeta = (E \rightarrow U, v_\bullet)$, define

$$(v, k) \circ (u, h) = (vu, kh).$$

We must check that this is still an allowed arrow. The base square commutes because the two smaller squares do. The section equation is $kh\sigma_a = k\tau_a u = v_a v u$. Finally,

$$C \simeq S \times_T D \simeq S \times_T (T \times_U E) \simeq S \times_U E,$$

so the composite square is cartesian. Composition is associative because scheme morphisms compose associatively in both entries of the pair. Composing with the stated identities changes neither entry. We have now checked all the category axioms.

Checkpoint. So far we have an ordinary category. There has been no gluing theorem and no assertion about a parameter space representing it.

Exercise 1.2. Let $C_0 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ be the constant genus-two family. Explain why the inclusion of the fiber at 0 defines an arrow in the total category, but has no inverse there. Why does this not contradict the forthcoming claim that each fiber category is a groupoid?

1.3 The category over schemes and its fibers

A *category over* $\text{Sch}/\mathbb{C}$ starts with a category and asks us to specify a functor from it to $\text{Sch}/\mathbb{C}$. It need not yet have pullbacks or satisfy any descent condition. Here define

$$p : \overline{\mathcal{M}}_{g,n} \longrightarrow \text{Sch}/\mathbb{C}, \quad p(\xi) = S, \quad p(u, h) = u.$$

The identity calculation gives $p(\text{id}_\xi) = \text{id}_S$, and the composition calculation gives $p((v, k) \circ (u, h)) = v \circ u$. Thus p is a covariant functor. If this is the functor previously called **Target**, then $p = \text{Target}$: it retains the target S of the structural morphism $C \rightarrow S$. Later, for stable maps, this word will still mean S , not the fixed target variety X . This is Fantechi's Section 3.2 notion of a category over the base [1].

Definition 1.3 (The fiber category). Fix S . The category $\overline{\mathcal{M}}_{g,n}(S)$ has all families over this exact scheme S as objects. Its arrows are the arrows of the total category lying over id_S . Thus an arrow is an S -isomorphism $h : C \xrightarrow{\sim} D$ satisfying $h\sigma_a = \tau_a$.

The word “fiber” here means a category, not the geometric fiber $C_{\bar{s}}$ of a scheme morphism. Objects in $\overline{\mathcal{M}}_{g,n}(S)$ can be entire varying families. For $S = \text{Spec } \mathbb{C}$, they are individual stable pointed curves and their marked isomorphisms. The requirement that arrows lie over the *identity* is important: an automorphism of S different from the identity would give an arrow in the total category, but not in this fiber category.

1.4 Pullbacks and cartesian arrows

1.4.1 Construct the changed family

Suppose we have a family ξ over S and a $\mathbb{C}$ -morphism $u : T \rightarrow S$. We are asked to produce its family over T . Set $C_T = T \times_S C$, let $\pi_T : C_T \rightarrow T$ be projection, and put $\sigma_{a,T} = (\text{id}_T, \sigma_a u)$. Then

$$\begin{array}{ccc} C_T = T \times_S C & \xrightarrow{q} & C \\ \pi_T \downarrow & & \downarrow \pi \\ T & \xrightarrow{u} & S \end{array}$$

is a cartesian square and $q\sigma_{a,T} = \sigma_a u$.

Properness, flatness, and finite presentation survive base change. So do the smoothness and disjointness conditions on the sections. The geometric fibers of π_T are geometric base changes of those of π ; connectedness, nodality, genus, and the component stability test are unchanged by extension of algebraically closed fields. Consequently $u^*\xi = (C_T \rightarrow T, \sigma_{\bullet,T})$ is an allowed object. The pair $(u, q) : u^*\xi \rightarrow \xi$ is an allowed arrow. These base-change statements are standard family-of-curves facts; see [4, Section 6.3.4, pp. 231–232].

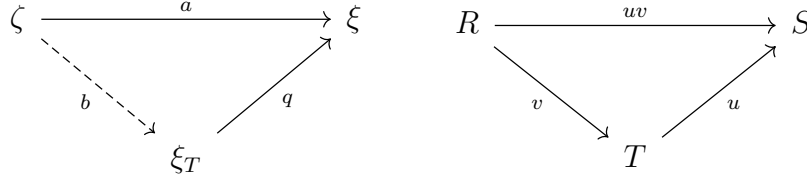
Example 1.4 (Pulling back the genus-two family). For the constant family $C_0 \times S \rightarrow S$, changing the base to T gives $C_0 \times T \rightarrow T$. Pullback to a point of S selects a fiber, while pullback to an infinitesimal scheme such as $\text{Spec } \mathbb{C}[\epsilon]/(\epsilon^2)$ also remembers infinitesimal parameter information. Pullback is thus more than substituting complex numbers into an equation.

1.4.2 The categorical universal property

The word “cartesian” has a second, related usage, for an arrow in a category over a base. Its definition starts with an arrow $q : \xi_T \rightarrow \xi$ over $u : T \rightarrow S$. The question is whether all arrows whose base maps factor through u factor uniquely through q in the corresponding way.

Definition 1.4 (Cartesian arrow). An arrow $q : \xi_T \rightarrow \xi$ over $u : T \rightarrow S$ is cartesian if for every object ζ over R , base map $v : R \rightarrow T$, and arrow $a : \zeta \rightarrow \xi$ over uv , there is a

unique arrow $b : \zeta \rightarrow \xi_T$ over v with $qb = a$.



The base factorization v is specified; it is not something to be chosen as part of the asserted uniqueness.

Let us verify this for the pullback just constructed. Write the total space of ζ as E , its projection as $\lambda : E \rightarrow R$, and the total-space map of a as $a_C : E \rightarrow C$. There is exactly one scheme map

$$b_C = (v\lambda, a_C) : E \longrightarrow T \times_S C$$

with those two projections, by the scheme fiber-product property. It respects the marked sections because both of its projections do. It remains to check that it defines a *moduli arrow*: since a is an arrow, $E \simeq R \times_S C$; on the other hand,

$$R \times_T (T \times_S C) \simeq R \times_S C.$$

Thus $E \rightarrow R \times_T C$ is an isomorphism, exactly the required cartesian-square condition for (v, b_C) . Uniqueness as a scheme map implies uniqueness as a moduli arrow. This proves the full universal property, including the requirement that the candidate factorization belongs to our category.

Exercise 1.3. If two pullback objects ξ_T, ξ'_T with arrows to ξ satisfy Definition 1.4, show that there is a unique isomorphism between them commuting with the arrows to ξ . Why is it incorrect to omit “commuting with the arrows” when asserting uniqueness for a genus-two curve with an involution?

1.5 Why the category is fibered in groupoids

A category over schemes is *fibered* if an object over S has a cartesian pullback arrow over every map $T \rightarrow S$. We have constructed one and checked the universal property. A *groupoid* is a category in which every arrow is invertible. In $\overline{\mathcal{M}}_{g,n}(S)$, the cartesian-square requirement over id_S says precisely that $h : C \rightarrow D$ is an isomorphism. Its inverse is over S and satisfies $h^{-1}\tau_a = \sigma_a$, so it too is a fiber arrow. Therefore all fibers are groupoids.

We have proved that $p : \overline{\mathcal{M}}_{g,n} \rightarrow \text{Sch}/\mathbb{C}$ is a *category fibered in groupoids*. In fact every arrow in this total category is cartesian, since its definition identifies its source with the appropriate pullback. Thus it also satisfies Fantechi's Definition 3.1 in Section 3.5 [1]: liftings with a fixed target and base map are unique up to a unique compatible isomorphism. Chosen iterated pullbacks need not be literally equal: $(uv)^*\xi$ and $v^*u^*\xi$ have their canonical isomorphism. All overlap equations below use those identifications, as Fantechi explains in Sections 4.1 and 6.3.

Checkpoint. Pullback explains how to move a known family to a new base. A stack needs the reverse operation too: recover global data from compatible local data. Neither part of that gluing assertion has been proved by merely checking the category axioms.

1.6 What the two stack conditions ask

Fix an étale cover $\{S_i \rightarrow S\}$. Write $S_{ij} = S_i \times_S S_j$ and $S_{ijk} = S_i \times_S S_j \times_S S_k$. These are the places where two or three descriptions of the same base point can be compared. For an open cover they are intersections. For a noninjective cover they are actual fiber products, with several possible comparisons even when there is only one index i .

Definition 1.5 (Fantechi's stack conditions). Start with a category fibered in groupoids $\mathcal{F}$ over our site. It is a stack if the following two requirements hold.

- (a) *Isomorphisms are a sheaf.* Given two objects ξ, η already over S , and local isomorphisms $a_i : \xi|_{S_i} \rightarrow \eta|_{S_i}$ agreeing over every S_{ij} , there is exactly one global isomorphism $a : \xi \rightarrow \eta$ whose restrictions are the a_i .
- (b) *Every object descent datum is effective.* Given objects ξ_i over S_i and isomorphisms $\alpha_{ij} : \xi_i|_{S_{ij}} \rightarrow \xi_j|_{S_{ij}}$ satisfying

$$\alpha_{jk} \circ \alpha_{ij} = \alpha_{ik} \quad \text{over } S_{ijk}, \tag{1.6.1}$$

there is an object ξ over S and isomorphisms $b_i : \xi|_{S_i} \rightarrow \xi_i$ such that

$$\alpha_{ij} = b_j \circ b_i^{-1} \quad \text{over } S_{ij}. \tag{1.6.2}$$

This is Fantechi [1, Sections 4.2–4.3, Definitions 4.1–4.3]. In the requested local `StacksForEverybody.tex`, the descent definition is labelled `def:4.1` (lines 207–211), the isomorphism-sheaf condition is `def:4.2` (lines 219–221), and the stack definition

is `def:4.3` (lines 225–227). The source-PDF markers place them on pp. 6, 6, and 7, respectively.

The first condition begins with global objects and asks us to glue *an arrow*. The second begins with local objects and asks us to build *a global object*. A list of isomorphism classes of local objects is insufficient for the second: the actual transition isomorphisms and their cocycle are part of the input. On a triple overlap, going from description i to j and then to k must agree with going directly from i to k .

For fixed ξ, η over S , the sheaf in the first condition is the assignment

$$(T \xrightarrow{v} S) \longmapsto \text{Isom}_T(v^*\xi, v^*\eta),$$

with restriction maps given by pullback. Each value is a *set of isomorphisms*, not a set of curves. The sheaf assertion means existence and uniqueness of gluing for this assignment on covers of every T . It makes no assertion here that this sheaf is a scheme; representability will be a further geometric property.

1.7 First stack condition: isomorphisms form a sheaf

Let $\xi = (C \xrightarrow{\pi} S, \sigma_\bullet)$ and $\eta = (D \xrightarrow{\rho} S, \tau_\bullet)$ be stable pointed families. Suppose $a_i : C_i = C \times_S S_i \xrightarrow{\sim} D_i = D \times_S S_i$ are S_i -isomorphisms preserving all labels, with $a_i|_{S_{ij}} = a_j|_{S_{ij}}$. We must construct exactly one S -isomorphism with these restrictions.

The theorem that glues the underlying maps. For schemes Y, Z and an étale cover $\{Y_i \rightarrow Y\}$, compatible maps $Y_i \rightarrow Z$ glue to a unique map $Y \rightarrow Z$. More generally this holds for fpqc covers: Vistoli [2, Proposition 4.31, pp. 91–92] uses the representable-sheaf theorem, and [11, Lemma 35.13.7, Tag 023Q] states that theorem precisely. On affine pieces it is faithfully flat descent of ring homomorphisms, including the equalizer $A \rightarrow B \rightrightarrows B \otimes_A B$; the theorem extends the affine calculation to scheme morphisms. This is our external morphism-descent input, not a consequence of the stack we are trying to prove exists.

Apply it to the correct covering. The maps $C_i \rightarrow C$ are an étale cover, by base change from $S_i \rightarrow S$. Compose a_i with $D_i \rightarrow D$ to get maps $C_i \rightarrow D$. Their overlaps agree by assumption, so the theorem gives a unique scheme map $a : C \rightarrow D$. The two maps from C to S given by ρa and π agree after this cover; uniqueness in the same theorem gives $\rho a = \pi$. Thus a is over S , and its actual pullback to S_i is a_i .

Recover invertibility. The inverse maps a_i^{-1} also agree on overlaps. Applying the theorem to the cover $\{D_i \rightarrow D\}$ gives $b : D \rightarrow C$. Both ba and id_C restrict to id_{C_i} , hence are equal; likewise $ab = \text{id}_D$. Consequently a is an isomorphism. We have not inferred invertibility merely from a bijection on fibers.

Recover the markings and uniqueness. For each marking index r , the maps $a\sigma_r, \tau_r : S \rightarrow D$ agree on all S_i , so they are equal. Hence the global isomorphism is an arrow in $\overline{\mathcal{M}}_{g,n}(S)$. Any other such arrow with the same local restrictions is equal as a scheme map to this one. This proves existence, the required properties, and uniqueness. The argument works over any test scheme $T \rightarrow S$, so it proves the sheaf condition in full.

Example 1.5 (Compatible genus-two isomorphisms). Let $S = \mathbb{A}^1$, with $U = S \setminus \{0\}$ and $V = S \setminus \{1\}$. Take $C = D = C_0 \times S$. The local maps $\iota \times \text{id}_U$ and $\iota \times \text{id}_V$ agree on $U \cap V$ and glue to $\iota \times \text{id}_S$. If instead we take the identity on U and ι on V , they disagree on the overlap and no global map has those restrictions. The sheaf condition never promises to glue incompatible arrows. This example illustrates the general proof; it does not prove the condition for other families or other covers.

Exercise 1.4. Why is it not enough to glue just the underlying maps and say “therefore they are isomorphisms”? Explain both a way to finish using their inverses and the separate check required by markings.

1.8 The geometric tool that makes object descent possible

We now need a theorem that produces a scheme from local schemes. Before applying it to our curves, we explain its input: a line bundle that supplies compatible projective coordinates. Merely knowing that every local curve admits *some* projective embedding does not supply compatible embeddings on overlaps. The particular line bundle below is determined by the pointed curve itself, so every transition isomorphism automatically acts on it.

Definition 1.6 (The line bundle supplied by a stable pointed family). Start with the family $(p : C \rightarrow S, \sigma_1, \dots, \sigma_n)$ of Definition 1.1. Each marking has an image $D_a = \sigma_a(S)$ in C . We form

$$D = D_1 + \dots + D_n, \quad L_C = \omega_{C/S}(D) = \omega_{C/S} \otimes \mathcal{O}_C(D).$$

Here $\omega_{C/S}$ is the *relative dualizing sheaf*: the line bundle playing the role of relative differential forms for a family whose fibers are allowed to have nodes. The notation (D) means tensoring with the line bundle associated to the indicated divisor.

This definition uses two results about families of schemes, which we state as external inputs. A section through the smooth locus of a separated family of curves is a relative effective Cartier divisor. Concretely, its image is locally cut out by one equation that is a non-zero-divisor, and this description remains a Cartier divisor after base change. One can check this in a relative smooth coordinate, where the section has equation $z = 0$; the general statement follows by étale localization. See Stacks Project, Section 44.3, Lemma 44.3.1, and Section 31.23, Lemma 31.23.8 [11]. Thus all the $\mathcal{O}_C(D_a)$ above are defined and commute with base change.

The second input is the relative duality theorem in its curve form: if $p : C \rightarrow S$ is proper, flat, and finitely presented, with Gorenstein geometric fibers of pure dimension one, then $\omega_{C/S}$ is invertible and, for every $b : T \rightarrow S$, there is a canonical isomorphism

$$\mathrm{pr}_C^* \omega_{C/S} \simeq \omega_{C \times_S T/T}. \quad (1.8.1)$$

These isomorphisms respect further base change and isomorphisms of families. This is a theorem about proper flat families; it does not require us to know that our moduli category is algebraic. A precise curve formulation is Stacks Project, Tag 0E6R, Lemma 109.19.3, with its duality references in Section 109.19 [11]. We use this duality statement as a black box.

Why does it apply? A smooth curve has regular local rings; a node is locally a hypersurface singularity, with completed local ring $\bar{k}[[x, y]]/(xy)$. These rings are Gorenstein: their dualizing modules have rank one and are free. Consequently the geometric fibers in our definition satisfy the theorem. On the smooth locus, $\omega_{C/S}$ agrees with relative differentials. On a single nodal fiber its sections can be described on the normalization as differentials with at most simple poles over each node, with the two residues summing to zero. This explains why ordinary regular differentials on a smooth curve need a replacement at a node.

Lemma 1.1 (Functorial relative ampleness). *For a stable pointed family, $L_C = \omega_{C/S}(D)$ is relatively ample over S . It commutes canonically with arbitrary base change. A pointed isomorphism $a : C \xrightarrow{\sim} C'$ over S therefore gives a canonical isomorphism $a^* L_{C'} \simeq L_C$, compatible with composition of pointed isomorphisms.*

Verification. The base-change assertion follows from (1.8.1) and from the relative Cartier divisor property of the sections. For ampleness, restrict to a geometric fiber and normalize any one of its irreducible components. If that normalization has genus h and r distinguished points, counting the branches of nodes and the markings, then

$$\deg L_C|_{\tilde{E}} = 2h - 2 + r > 0 \quad (1.8.2)$$

by stability. Positivity on every component implies ampleness on the fiber. The precise curve criterion is Stacks Project, Tag 0B5Y, Lemma 33.44.15; degrees are unchanged on normalization by Lemma 33.44.4. For a proper family, ampleness on a fiber implies relative ampleness on a neighborhood of its base point (Tag 0D2S, Lemma 37.50.3). These neighborhoods cover S , which proves relative ampleness. The final assertion follows because an isomorphism carries each labelled section to the corresponding labelled section, and the duality identifications are functorial. $\square$

In ordinary language, every stable pointed family comes with the *same rule* for choosing an ample line bundle. We have not added a line bundle to the objects of $\overline{\mathcal{M}}_{g,n}$: we have constructed one from data already present. For a smooth unpointed genus-two curve C , this rule is simply $L_C = \omega_C$, of degree 2. For a genus-two curve made from two smooth elliptic curves joined at one node, its degree on each component is 1. The positivity check works on both components even though the curve is singular.

Theorem 1.1 (Effective descent with a compatible ample line bundle). *Let $\{S_i \rightarrow S\}$ be an étale cover of a scheme. Suppose that for every i we are given a proper, flat, finitely presented morphism $p_i : Y_i \rightarrow S_i$ and a relatively ample invertible sheaf L_i on Y_i . Suppose also that there are isomorphisms*

$$a_{ij} : Y_i|_{S_{ij}} \xrightarrow{\sim} Y_j|_{S_{ij}}, \quad \lambda_{ij} : a_{ij}^*(L_j|_{S_{ij}}) \xrightarrow{\sim} L_i|_{S_{ij}},$$

where $S_{ij} = S_i \times_S S_j$. Assume the scheme isomorphisms satisfy $a_{ik} = a_{jk} \circ a_{ij}$ on S_{ijk} , and the line-bundle isomorphisms satisfy

$$\lambda_{ik} = \lambda_{ij} \circ a_{ij}^* \lambda_{jk} \quad \text{on } S_{ijk}, \tag{1.8.3}$$

using the canonical identifications of iterated pullbacks. Then there are a proper, flat, finitely presented scheme $Y \rightarrow S$, a relatively ample invertible sheaf L on Y , and isomorphisms $(Y, L)|_{S_i} \simeq (Y_i, L_i)$ recovering the prescribed descent data. The same statement holds for fpqc covers.

Source and proof strategy. The scheme-and-line-bundle effectivity theorem is SGA 1, Exposé VIII, Proposition 7.8, p. 174 of the recomposed edition [3]. Its argument descends a graded algebra and uses relative Proj. Vistoli's later exposition [2], Section 4.3.3, Theorem 4.38 and proof, pp. 96–103, gives the detailed version for a functorial ample bundle on a class of proper flat finitely presented families. His theorem applies this method to a whole moduli problem; the statement above isolates the descent datum that we need. Neither source says that arbitrary projective schemes descend effectively without compatible polarization data.

Our guided explanation of the construction. The input is local spaces together with instructions for identifying them and their line bundles on overlaps. The output must be one scheme over S , with identifications that reproduce precisely those instructions. We describe the *étale* case used in these notes.

First work over an affine open of S . Refine the cover by affine open subschemes of its members. Because étale maps are open and S is quasi-compact, finitely many of their images cover S . Their disjoint union U is affine, and $U \rightarrow S$ is faithfully flat and quasi-compact. The map need not be finite: it is the *list of affine covering schemes* that is finite. All the original data restrict to U and $U \times_S U$. After carrying out descent there, the uniqueness of morphism descent identifies the answer with the data on every original S_i . Answers over different affine opens of S glue by ordinary Zariski gluing. This reduction justifies using one uniform power of a line bundle in the following steps.

- (a) *Turn the local families into closed subschemes of projective bundles.* Write $p_U : Y_U \rightarrow U$ and L_U for the disjoint union of the refined data. Relative Serre vanishing and the relative very-ampleness theorem allow us to choose $m > 0$ such that $L_U^{\otimes m}$ is relatively very ample and $H^j(Y_u, L_u^{\otimes m}) = 0$ for every $j > 0$ on every fiber. Cohomology and base change then makes

$$E_U = (p_U)_* L_U^{\otimes m}$$

a finite locally free sheaf, with formation commuting with base change. These are substantial scheme-theoretic inputs, not consequences of the stack axioms. The cohomology statement, including arbitrary bases, is Vistoli, Proposition 4.37, pp. 95–96; the passage to a sufficiently large power occurs in the proof of Theorem 4.38. The evaluation map $p_U^* E_U \rightarrow L_U^{\otimes m}$ gives a closed immersion

$$Y_U \hookrightarrow \mathbb{P}_U(E_U) := \mathrm{Proj}_U(\mathrm{Sym}^\bullet E_U).$$

Here relative Proj is the construction that turns a graded algebra over a base into a projective scheme over that base.

- (b) *Descend the projective coordinates.* The isomorphisms a_{ij} and λ_{ij} induce compatible isomorphisms between the pullbacks of E_U on $U \times_S U$. Equation (1.8.3) gives their cocycle condition. Faithfully flat descent for quasi-coherent modules now constructs a sheaf E on S whose pullback is E_U . Local freeness and finite rank can be checked faithfully flat locally, so E is a vector bundle. We use Vistoli, Theorem 4.23, pp. 85–90, for this descent theorem. It is the algebraic input that replaces elementary gluing of vector bundles on open subsets.

Thus the projective bundles are all pullbacks of the single scheme $\mathbb{P}_S(E)$. Functoriality of evaluation ensures that the given isomorphisms identify the embedded copies of Y_U in this common projective bundle. This is the step for which compatible line bundles were essential.

- (c) *Descend the equations.* Let $\mathcal{I}_U \subset \mathcal{O}_{\mathbb{P}_U(E_U)}$ be the ideal sheaf of the embedded Y_U . The compatibility just checked gives this ideal sheaf a descent datum over the fpqc cover $\mathbb{P}_U(E_U) \rightarrow \mathbb{P}_S(E)$. Quasi-coherent descent constructs an ideal sheaf $\mathcal{I} \subset \mathcal{O}_{\mathbb{P}_S(E)}$; both its inclusion and its closure under multiplication can be checked after the cover. Let $Y \subset \mathbb{P}_S(E)$ be the closed subscheme cut out by $\mathcal{I}$. Formation of this closed subscheme commutes with flat base change, so $Y \times_S U \simeq Y_U$ with the *specified* transition identifications. This is the projective-bundle form of the graded-algebra/Proj construction in SGA 1.
- (d) *Recover the line bundle and check the family.* Now that Y exists, the L_i are line bundles on a cover of Y with compatible transition maps. Quasi-coherent descent constructs L ; invertibility is checked on that cover. Properness, flatness, and finite presentation of $Y \rightarrow S$ follow from their fpqc locality on the base (Stacks Project, Tag 02L1, Tag 02L2, Tag 02L0, Lemmas 35.23.16, 35.23.17, 35.23.15). Relative ampleness of L can be checked on geometric fibers and then near each base point using the proper-family ampleness theorem above.

This explains why the theorem returns a *scheme*. It does not begin by assuming that the sheaf quotient of the Y_i is a scheme. The vector bundle and the descended equations construct that scheme. The theorem has not yet supplied markings or a map to a target; we will descend those additional morphisms after applying it.

Exercise 1.5. Suppose the local pointed families have transition isomorphisms a_{ij} satisfying the cocycle condition. Explain why choosing $L_i = \omega_{C_i/S_i}(\sum_a \sigma_{a,i})$ supplies the λ_{ij} and equation (1.8.3) without an additional choice. Why would choosing an unrelated ample line bundle on each C_i not by itself do this?

1.9 Second stack condition: descent data are effective

The input now differs from that of the preceding sheaf proof. There need not be a known global curve. We are given an étale cover $\{S_i \rightarrow S\}$, families

$$(C_i \xrightarrow{\pi_i} S_i, \sigma_{1,i}, \dots, \sigma_{n,i}),$$

and isomorphisms

$$\alpha_{ij} : C_i \times_{S_i} S_{ij} \xrightarrow{\sim} C_j \times_{S_j} S_{ij}$$

over S_{ij} taking every $\sigma_{a,i}$ to $\sigma_{a,j}$. On S_{ijk} , we are given the equality $\alpha_{jk}\alpha_{ij} = \alpha_{ik}$ of scheme maps. That last requirement means the following triangle commutes:

$$\begin{array}{ccc} C_i|_{S_{ijk}} & \xrightarrow{\alpha_{ij}} & C_j|_{S_{ijk}} \\ & \searrow \alpha_{ik} & \downarrow \alpha_{jk} \\ & & C_k|_{S_{ijk}} \end{array}$$

All three vertices are curves over the *same* triple overlap. For a nontrivial étale cover, one cannot replace these fiber products by informal intersections.

Step 1: provide compatible ample bundles. On C_i form $L_i = \omega_{C_i/S_i}(\sum_{a=1}^n \sigma_{a,i})$. The preceding descent discussion establishes that it is invertible and relatively ample, commutes with base change, and is carried to the corresponding bundle by an isomorphism of pointed families. In particular $\alpha_{ij}^* L_j \simeq L_i$ on the overlap. These are the canonical pullback identifications, so their composites on triple overlaps agree. There is no arbitrary scalar choice of transition maps on the line bundles.

Step 2: construct the global scheme. Apply the polarized scheme-descent theorem stated above to (C_i, L_i, α_{ij}) . All its hypotheses have just been checked: the cover is faithfully flat locally of finite presentation after the stated refinements, each $C_i \rightarrow S_i$ is proper, flat, and finitely presented, the bundles are relatively ample, and both the scheme and bundle transition maps satisfy their cocycles. We obtain a scheme $C \rightarrow S$, a relatively ample bundle L , and isomorphisms

$$b_i : C \times_S S_i \xrightarrow{\sim} C_i$$

whose transition maps are the given α_{ij} . These identifications are part of the output: effectiveness asks for them, not just for some global curve with isomorphic fibers.

Properness, finite presentation, and flatness of $C \rightarrow S$ can be checked after the faithfully flat cover, so they hold because they hold for every π_i . This uses the descent-of-properties theorem [2, Proposition 1.15, pp. 11–12], or precisely Tag 02L0, Tag 02L1, and Tag 02L2 in [11].

Step 3: descend the sections. The local sections, transported by b_i^{-1} and then projected to C , give maps $s_{a,i} : S_i \rightarrow C$. They agree on S_{ij} because α_{ij} preserves marking a . The morphism-sheaf theorem gives a unique $\sigma_a : S \rightarrow C$ with these restrictions. The equation $\pi\sigma_a = \text{id}_S$ holds after the cover, hence globally. Thus we have recovered genuine sections, including over nonreduced parameter schemes.

Their images lie in the smooth locus because this assertion is local after étale base change. They are disjoint: a point in an intersection would lift after the surjective cover to an intersection of the corresponding local sections, which does not exist. Equivalently their fiber product is empty after a faithfully flat cover, so it was empty to start with.

Step 4: check the geometric fibers and stability. Take any geometric point $\bar{s} : \text{Spec } \Omega \rightarrow S$. After passing to an algebraically closed extension if necessary, it lifts to some S_i . The geometric fiber of C at that lifted point is, by b_i , the corresponding fiber of C_i . It is connected, reduced, nodal, has genus g , has distinct smooth markings, and meets the component stability inequalities. These geometric assertions are unchanged by algebraically closed field extension; hence they hold for $C_{\bar{s}}$. We have verified every part of Definition 1.1. The recovered L is also canonically the bundle $\omega_{C/S}(\sum \sigma_a)$, since that identification holds locally and the compatible bundle maps descend.

Step 5: check the prescribed descent datum. By construction $\alpha_{ij} = b_j b_i^{-1}$. The b_i preserve all recovered sections, so they are isomorphisms in the fiber categories, not just isomorphisms of unmarked schemes. We have therefore produced exactly the object and identifications required by Fantechi's Definition 4.1.

If another family $C' \rightarrow S$ with identifications b'_i has the same descent datum, the local maps $b_i^{-1} b'_i$ agree on overlaps. The first stack condition glues them to a unique global isomorphism compatible with all identifications. This is the appropriate uniqueness statement; the object need not have a trivial automorphism group.

1.9.1 A nontrivial genus-two descent datum

Let $S = \text{Spec } \mathbb{C}[t, t^{-1}]$ and let $U = \text{Spec } \mathbb{C}[u, u^{-1}] \rightarrow S$ be $t = u^2$. Over U take the constant family $C_0 \times U$. The overlap $U \times_S U$ has two components, specified by $u_2 = u_1$ and $u_2 = -u_1$. Define the transition map to be the identity on the first component and ι on the second. In affine coordinates it is

$$(x, y_1) \longmapsto (x, y_2), \quad y_2 = (u_1/u_2)y_1.$$

On a triple overlap the factors multiply as $(u_2/u_3)(u_1/u_2) = u_1/u_3$, so the cocycle holds. The data are a family and a rule comparing its two possible square-root descriptions, even though the cover has only one member.

Here the descended family can be written explicitly as the smooth projective double cover with affine equation

$$z^2 = t(x^6 - 1).$$

To specify the completion over the whole base, use the section $t(X^6 - Z^6)$ of $\mathcal{O}_{\mathbb{P}_S^1}(6)$ and the usual double-cover algebra $\mathcal{O}_{\mathbb{P}_S^1} \oplus \mathcal{O}_{\mathbb{P}_S^1}(-3)$, whose multiplication is determined by that section. Its relative spectrum is finite over $\mathbb{P}_S^1$, hence proper over S . After the étale cover it is $C_0 \times U$, so it is flat, smooth of relative dimension one, and genus two. The local identification is $y = z/u$; these identifications have exactly the stated transition maps. The section $(x, z) = (1, 0)$ can be included if we want the one-pointed version.

Pullback along $T \rightarrow S$ replaces t by its invertible pullback function on T . In particular the original square-root cover makes the family constant. The example shows why transition maps are real information: locally constant descriptions can carry a nontrivial gluing rule. Our general descent theorem, not this equation, was what proved effectiveness for every stable-curve datum.

Exercise 1.6. For this one-map cover, explain why the overlap still contains more than the diagonal. Check the cocycle when the first comparison and the second comparison are both sign changes. Compute the transition on the local expression $y = z/u$.

Theorem 1.2 (The stable pointed-curve stack). *For $g, n \geq 0$ with $2g - 2 + n > 0$, the category of stable n -pointed genus- g curves defined above, with its projection to $\text{Sch}/\mathbb{C}$, is a stack for the étale topology.*

Proof. Identities and composition make it a category. The projection is a functor. Base change gives cartesian liftings and all fibers are groupoids. Compatible isomorphisms glue uniquely by Section 1.7. Every object descent datum is effective by Section 1.9. These are exactly the hypotheses and the two conclusions required by Definition 1.5. $\square$

1.10 Properties

The stack proof gives an answer to a gluing question: compatible local families, and compatible local isomorphisms, come from global ones. We now ask geometric questions about that stack. These require additional theorems. Throughout this section $g, n \geq 0$, $2g - 2 + n > 0$, and $\overline{\mathcal{M}}_{g,n}$ is the stack over $\mathbb{C}$ just constructed.

1.10.1 What the geometric words ask us to construct

Definition 1.7 (Representability and an atlas). Start with a morphism $F : \mathcal{Y} \rightarrow \mathcal{Z}$ of stacks. To test whether it is *representable by algebraic spaces*, choose any scheme S and any object $z \in \mathcal{Z}(S)$, equivalently a map $S \rightarrow \mathcal{Z}$, and form

$$\begin{array}{ccc} S \times_{\mathcal{Z}} \mathcal{Y} & \longrightarrow & \mathcal{Y} \\ \downarrow & & \downarrow F \\ S & \xrightarrow{z} & \mathcal{Z}. \end{array}$$

The requirement is that the fiber product be an algebraic space. An algebraic space is an étale sheaf which has a representable, surjective étale presentation by a scheme; it allows étale coordinate charts even when those charts do not glue to a scheme by open immersions. In the applications below these particular fiber products will actually be schemes.

If F is representable, saying that it is smooth, étale, or surjective means that the resulting maps of algebraic spaces to S have that property for every such test. An *atlas* is a representable surjective map $U \rightarrow \mathcal{Z}$ from a scheme, with the specified smoothness or étaleness condition.

For example, a map $U \rightarrow \overline{\mathcal{M}}_{g,n}$ is a family of stable pointed curves over U . Given another family C/S , a point of $S \times_{\overline{\mathcal{M}}_{g,n}} U$ consists of a parameter u in U and an *isomorphism* from the curve selected by u to the corresponding fiber of C/S . The isomorphism is part of the data. The fiber product does not merely record that two isomorphism classes agree.

Definition 1.8 (Algebraic, Artin, and Deligne–Mumford). Start with a stack $\mathcal{Z}$ on the étale site of schemes over $\mathbb{C}$. In these notes, it is an *algebraic stack*, also called an *Artin stack*, if its diagonal is representable by algebraic spaces and it has a smooth atlas $U \rightarrow \mathcal{Z}$ from a scheme. It is a *Deligne–Mumford stack* if it has an étale atlas. Thus every Deligne–Mumford stack is an algebraic stack in our convention, since étale morphisms are smooth.

The diagonal has a concrete interpretation here. Pulling $\Delta : \overline{\mathcal{M}}_{g,n} \rightarrow \overline{\mathcal{M}}_{g,n} \times \overline{\mathcal{M}}_{g,n}$ back by two families $C/S, D/S$ gives the sheaf $\text{Isom}_S(C, D)$, with the markings understood. The earlier sheaf proof showed that this functor is a sheaf. Representability of the diagonal asks for more: this sheaf must itself be an algebraic space. An atlas asks for still more: enough families must be parametrized by schemes in a smooth or étale way. For these conventions and their relation, see Alper [4, Definitions 4.1.3–4.1.6

and Theorem 4.2.1, pp. 105–106, 118–119]. Some older sources, including Deligne–Mumford, use “algebraic stack” for what we now call a Deligne–Mumford stack.

1.10.2 Algebraicity and finite type

Proposition 1.1. *The stack $\overline{\mathcal{M}}_{g,n}$ is algebraic and of finite type over $\mathbb{C}$. Its diagonal is affine and of finite type. It has a smooth surjective atlas $H \rightarrow \overline{\mathcal{M}}_{g,n}$, where H is a scheme of finite type parametrizing suitable embedded stable pointed curves.*

Original source and proof strategy. Deligne–Mumford [5, Section 1, pp. 77–78, and Proposition (5.1), p. 104] construct the unpointed stack using tricanonical Hilbert schemes. Knudsen [6, Theorem 2.7, p. 179] proves the pointed theorem over $\text{Spec } \mathbb{Z}$, using the unpointed result and induction in the number of markings. We give a direct Hilbert-atlas argument for pointed stable curves. Its embedding bound is Alper [4, Proposition 6.3.23, p. 231]; its smooth-atlas proof is the embedding-lifting argument of [4, Theorem 6.4.6, pp. 238–240], applied here directly to our stable category. Although that theorem is written for a larger category of curves, we do not construct that larger stack. Our parameter scheme retains some extra choices of embedding; these choices are allowed in an atlas.

Our guided proof. The data already available are arbitrary stable pointed families $(\pi : C \rightarrow S, \sigma_1, \dots, \sigma_n)$. We must construct one scheme containing enough such families and show that its map to our stack is representable, smooth, and surjective.

Step 1: use a canonical line bundle to bound embeddings. Put

$$L_C = \omega_{C/S} \left(\sum_{a=1}^n \sigma_a \right).$$

We use the following *external embedding theorem*: for a stable pointed family, $L_C^{\otimes 3}$ is relatively very ample, $E_C = \pi_* L_C^{\otimes 3}$ is locally free of rank $r = 5g - 5 + 3n$, and formation of this vector bundle commutes with arbitrary base change. These conclusions follow from the fiberwise vanishing and very ampleness theorem together with cohomology and base change; see Alper [4, Exercise 6.3.18 and Proposition 6.3.23, pp. 230–231]. The fiberwise unpointed theorem is proved in Deligne–Mumford [5, Theorem (1.2) and its corollary, pp. 77–78]. We take the pointed very ampleness result as an input rather than redo its separation-of-points-and-tangents proof.

The rank and Hilbert polynomial have a useful numerical explanation. On any geometric fiber, let $\ell = 2g - 2 + n > 0$. Then $\deg L_C = \ell$ and $\deg L_C^{\otimes 3} = 3\ell$. Vanishing of $H^1(C, L_C^{\otimes 3})$ and Riemann–Roch give

$$r = 3\ell + 1 - g = 5g - 5 + 3n, \quad P(m) = \chi(C, L_C^{\otimes 3m}) = 3\ell m + 1 - g.$$

These numbers depend only on g, n , not on the particular curve. The evaluation of sections gives a closed immersion $C \hookrightarrow \mathbb{P}_S(E_C)$, where our convention for $\mathbb{P}(E_C)$ parametrizes one-dimensional quotients of E_C . Locally on S , choosing a basis of E_C identifies the ambient bundle with $\mathbb{P}_S^{r-1}$.

Step 2: represent the diagonal. Take two stable pointed families $C/S, D/S$. After trivializing E_C, E_D locally over S , their canonical embeddings place them in the same projective space. An isomorphism $C \rightarrow D$ induces an isomorphism of their canonical line bundles, hence a unique projective linear transformation of these complete embeddings. Conversely, a projective transformation carrying the embedded curve and every marking to the corresponding data restricts to an isomorphism of pointed families.

Thus $\text{Isom}_S(C, D)$ is locally represented by a closed subscheme of $\text{PGL}_{r,S}$: the transformation must send one Hilbert point to the other, and must send each marked point to its partner. Equality of Hilbert points is closed because the Hilbert scheme is separated; the point conditions are closed because projective space is separated. These are finitely presented conditions, since the Hilbert scheme and projective space are of finite presentation over $\mathbb{C}$. The local representing schemes glue by descent for affine schemes, so $\text{Isom}_S(C, D)$ is an affine scheme of finite presentation over S . This proves the claimed representability and finiteness of the diagonal. It also implies that a map from any scheme to our stack is representable: its test fiber products are obtained by base changing that diagonal. Notice how much stronger this is than merely saying that $\text{Isom}_S(C, D)$ is a sheaf.

Step 3: choose a finite type parameter scheme. The Hilbert scheme theorem is an external input: closed subschemes of $\mathbb{P}_S^{r-1}$ flat and finitely presented over S , with fiber Hilbert polynomial P , are represented by a projective scheme $\text{Hilb}^P(\mathbb{P}_{\mathbb{C}}^{r-1})$. See Alper [4, Theorem 2.1.2, p. 48]. Adding n points and requiring each to lie on the universal curve gives a closed incidence scheme in

$$\text{Hilb}^P(\mathbb{P}_{\mathbb{C}}^{r-1}) \times (\mathbb{P}_{\mathbb{C}}^{r-1})^n.$$

Inside it, take the open locus H where the pointed fibers are stable and $H^1(C_s, \mathcal{O}_{C_s}(1)) = 0$. Openness of stability is Alper [4, Proposition 6.3.24, p. 232]; the vanishing locus is open by upper semicontinuity of cohomology. Both are substantial standard inputs about projective flat families. The Hilbert polynomial already fixes the arithmetic genus to be g . Since the incidence scheme is of finite type over the noetherian field $\mathbb{C}$, its open subscheme H is of finite type.

The universal family defines $a : H \rightarrow \overline{\mathcal{M}}_{g,n}$ by forgetting the embedding. It is representable by Step 2. It is surjective because any stable family is, locally on its base,

embedded by L_C^3 as in Step 1, and these embeddings satisfy the defining conditions of H . We deliberately allow other embeddings in H as well; being an atlas does not require a unique embedding for each curve.

Step 4: why forgetting the embedding is smooth. Smoothness says that an embedding can be chosen compatibly when a family undergoes a small infinitesimal variation. We apply the *infinitesimal criterion for smoothness* to the representable map a , following Alper [4, Theorem 4.7.1, pp. 144–145; proof of Theorem 6.4.6, pp. 238–240]. The finite-presentation hypothesis of the criterion also holds. For a test family C/S , the scheme $S \times_{\overline{\mathcal{M}}_{g,n}} H$ is the isomorphism scheme between the two families over $S \times H$: the pullback of C/S and the pullback of the universal family over H . Step 2 makes this scheme finitely presented over $S \times H$, and $S \times H$ is finitely presented over S . Thus a is representable and locally of finite presentation before we use the lifting calculation.

Let $A' \twoheadrightarrow A$ be a small extension of local Artinian $\mathbb{C}$ -algebras, with residue field K and kernel J killed by the maximal ideal of A' . We are given a stable pointed family C'/A' and an embedding of its restriction $C = C' \times_{A'} A$ in $\mathbb{P}_A^{r-1}$, representing a point of $H(A)$. We must lift that embedding to C' :

$$\begin{array}{ccc} C & \hookrightarrow & \mathbb{P}_A^{r-1} \\ \downarrow & & \downarrow \\ C' & \hookrightarrow & \mathbb{P}_{A'}^{r-1} \end{array}$$

The embedding on C is given by $M = \mathcal{O}_C(1)$ and its r coordinate sections.

A line bundle on C lifts to C' because its lifting obstruction lies in $H^2(C_K, \mathcal{O}_{C_K} \otimes_K J) = 0$. This elementary line-bundle lifting theorem follows from $1 + \mathcal{J} \rightarrow \mathcal{O}_{C'}^* \rightarrow \mathcal{O}_C^*$ for the square-zero ideal $\mathcal{J}$; see Alper [4, Proposition C.2.11 and Remark C.2.12, p. 602]. Choose a lift M' . The restriction sequence is

$$0 \longrightarrow M_K \otimes_K J \longrightarrow M' \longrightarrow M \longrightarrow 0.$$

The defining vanishing $H^1(C_K, M_K) = 0$ implies that every coordinate section of M lifts to M' . The lifted sections still generate M' by Nakayama's lemma. They define a morphism $C' \rightarrow \mathbb{P}_{A'}^{r-1}$ lifting the given embedding.

It is a closed immersion. Indeed it is proper, its only geometric fiber is a closed immersion and hence it is quasi-finite, so it is finite. The cokernel of the map from the ambient structure sheaf to its finite pushforward restricts to zero over A and therefore vanishes by Nakayama's lemma. The resulting embedded family is flat over A' because it is the given family C' . Its marked sections were already given and acquire their

images under this embedding. Its special fiber lies in H , so the lift also lies in the open locus H . This completes the required lifting. The infinitesimal criterion now proves that a is smooth.

We have constructed a representable smooth surjective map from the scheme H to $\overline{\mathcal{M}}_{g,n}$, and have represented the diagonal. These are exactly the requirements for algebraicity. Finally, finite type of H proves that $\overline{\mathcal{M}}_{g,n}$ is quasi-compact and locally of finite type over $\mathbb{C}$, which is the asserted finite type property.

Example 1.6 (Genus two and extra embedding choices). For $(g, n) = (2, 0)$ we have $\ell = 2$, $r = 5$, and $P(m) = 6m - 1$. Every stable genus-two curve has a tricanonical embedding in $\mathbb{P}^4$. Thus the Hilbert scheme for that one polynomial already contains embeddings of every stable genus-two curve. The atlas retains choices of an embedding and coordinates; $\overline{\mathcal{M}}_2$ retains the abstract curve and its isomorphisms. Extra parameters in an atlas are harmless: a smooth atlas need not have relative dimension zero.

1.10.3 Why the diagonal is unramified

Proposition 1.2. *Under the same hypotheses, $\overline{\mathcal{M}}_{g,n}$ is Deligne–Mumford. Every geometric stable pointed curve has a finite reduced automorphism group scheme.*

Original source and proof strategy. Deligne–Mumford [5, Lemma (1.4), pp. 80–81, Theorem (1.11), p. 84, and Proposition (5.1), p. 104] eliminate infinitesimal automorphisms and construct the unpointed Deligne–Mumford stack. Knudsen’s pointed theorem is [6, Theorem 2.7, p. 179]. The pointed calculation is explained by Alper [4, Proposition 6.3.26, pp. 232–235, and Theorem 6.4.16, p. 241]. The key geometric fact is that every component of the normalization has enough special points to eliminate regular vector fields fixing those points.

Our guided proof. We already have algebraicity. The additional construction we need is an étale atlas. We use the following *Deligne–Mumford criterion* as an external theorem: an algebraic stack has an étale atlas if and only if its diagonal is unramified, equivalently if every geometric stabilizer is discrete and reduced. In our finite type situation these stabilizers will be finite. See Alper [4, Theorem 4.6.4, p. 143].

Fix an algebraically closed extension $K/\mathbb{C}$ and a stable pointed curve $(C, p_1, \dots, p_n)$ over K . An *infinitesimal automorphism* starts with the constant family over $K[\epsilon]/(\epsilon^2)$ and is an automorphism that restricts to the identity when $\epsilon = 0$. Locally on functions it has the form $a \mapsto a + \epsilon D(a)$, where the multiplication rule forces D to be a derivation. Fixing a marked point forces the vector field D to vanish there. Thus the tangent space

of the automorphism scheme at its identity is

$$H^0(C, \mathcal{H}om(\Omega_C, \mathcal{O}_C(-\sum p_a))).$$

Let $\nu : \tilde{C} \rightarrow C$ be the normalization. On a nodal curve a regular vector field lifts to the normalization and vanishes at each branch over every node. This can be checked locally in $K[[x, y]]/(xy)$ by applying a derivation to $xy = 0$; see Alper [4, Exercise 6.2.22, p. 221]. Write C_v for the smooth connected components of $\tilde{C}$, h_v for their genera, and B_v for the marked points and branches of nodes on C_v . If $m_v = \deg B_v$, stability says

$$2h_v - 2 + m_v > 0.$$

Consequently

$$\deg T_{C_v}(-B_v) = 2 - 2h_v - m_v < 0.$$

A line bundle of negative degree on a smooth projective connected curve has no nonzero section. Every component vector field therefore vanishes, so the infinitesimal automorphism vanishes.

The automorphism scheme is an affine group scheme of finite type by the preceding diagonal calculation. Its zero tangent space at the identity implies that the identity is an isolated reduced point: in its noetherian local ring the equality $\mathfrak{m}/\mathfrak{m}^2 = 0$ implies $\mathfrak{m} = 0$ by Nakayama's lemma. Translation gives the same conclusion at every geometric point. A zero-dimensional reduced scheme of finite type over K has finitely many points. Thus the stabilizer is finite and reduced, and the Deligne–Mumford criterion applies.

In ordinary language, there can be discrete symmetries of the curve, but there is no continuously varying infinitesimal symmetry. For a genus-two hyperelliptic curve, the involution remains an automorphism; it does not prevent the stack from being Deligne–Mumford. It does prevent us from identifying that moduli problem with a scheme which has the same fiber groupoids.

Exercise 1.7. Check the three cases $h_v = 0$, $h_v = 1$, and $h_v \geq 2$ in the inequality above. Why would a smooth unmarked elliptic curve fail this argument? Explain why “finite automorphism groups” and “trivial automorphism groups” are different requirements.

1.10.4 Properness and the meaning of a limit

Proposition 1.3. *The Deligne–Mumford stack $\overline{\mathcal{M}}_{g,n}$ is separated and proper over $\mathbb{C}$.*

Original source and proof strategy. Deligne–Mumford [5, Section 2 and Theorem (5.2), pp. 104–105] prove the unpointed properness theorem using stable reduction and uniqueness of stable limits. Knudsen [6, Theorem 2.7, p. 179] proves the pointed version. A later detailed explanation in characteristic zero is Alper [4, Theorem 6.5.1 and Section 6.5.1, pp. 242–247; Proposition 6.5.20 and Theorem 6.5.23, pp. 251–252]. Alper’s written uniqueness proof is unpointed; the marked extension is assigned in Exercise 6.5.22. We take the full pointed existence and uniqueness statements from the established pointed theorem as external inputs. We do not claim to have supplied a new proof of stable reduction here.

Our guided proof. For an algebraic stack, *separated* means that its diagonal is proper. In terms of curves this is a statement about extending *isomorphisms* between families. *Proper over $\mathbb{C}$* means separated, of finite type, and universally closed over $\mathbb{C}$. The last condition is a completeness condition: limits cannot escape after any change of base. To test it effectively we use the stack valuative criterion [4, Theorem 4.8.7, pp. 149–150].

Here is the data in that criterion. Let R be a discrete valuation ring over $\mathbb{C}$, let K be its fraction field, and let $(C_K, p_1, \dots, p_n)$ be a stable pointed curve over K . Think of $\text{Spec } K$ as the punctured base and of the closed point of $\text{Spec } R$ as the missing limit. The *pointed stable reduction theorem* supplies, after a finite extension of the fraction field and a corresponding extension of discrete valuation rings $R \subset R'$, a stable pointed family $C' \rightarrow \text{Spec } R'$ whose generic fiber is the prescribed base change of C_K .

The second external input is *uniqueness of stable limits*: if C/R and D/R are two stable pointed families, then every specified pointed isomorphism $C_K \rightarrow D_K$ extends to a unique pointed isomorphism $C \rightarrow D$. The word “specified” matters. An object may have automorphisms; uniqueness concerns extension of one given generic isomorphism, not uniqueness of all possible isomorphisms between two limits.

Apply uniqueness to the valuative criterion for the representable diagonal. The diagonal is already affine and of finite type, hence quasi-compact and separated. The criterion proves that it is proper, so $\overline{\mathcal{M}}_{g,n}$ is separated. In particular its diagonal is finite, since an affine proper morphism is finite. Apply existence to the valuative criterion for the finite type stack $\overline{\mathcal{M}}_{g,n} \rightarrow \text{Spec } \mathbb{C}$. The permitted finite base extension in this criterion is exactly what stable reduction provides. The criterion proves universal closedness and hence properness. The hypotheses of the criterion have all been supplied: algebraicity, finite type, and quasi-separatedness came from the Hilbert presentation, and separatedness was just checked. Local noetherianness follows from finite type over $\mathbb{C}$.

Why does stable reduction involve geometry beyond descent? An étale cover sees every point of the base. The inclusion $\text{Spec } K \hookrightarrow \text{Spec } R$ misses the closed point, so

the stack condition cannot manufacture a curve over that point. Stable reduction is the theorem that does so, using the particular geometry of nodal stable curves.

Example 1.7 (A genus-two stable boundary point). Let E_1, E_2 be smooth elliptic curves and identify one point on each to form $C = E_1 \cup_q E_2$. Its arithmetic genus is two, and each elliptic component has one node as a special point, so C is stable. A smoothing of this node has local equation $xy = t$; the smooth fibers have genus two. The stable fiber at $t = 0$ belongs to $\overline{\mathcal{M}}_2$, while it does not belong to the open smooth-curve stack $\mathcal{M}_2$. The deformation theorem below supplies such smoothings. Uniqueness of stable limits shows that this punctured family cannot instead acquire a smooth special fiber after finite base extension. This illustrates why $\mathcal{M}_2$ is not proper and why the bar in $\overline{\mathcal{M}}_2$ matters.

1.10.5 Smoothness and actual dimension

Proposition 1.4. *The stack $\overline{\mathcal{M}}_{g,n}$ is smooth over $\mathbb{C}$ and has pure dimension $3g - 3 + n$.*

Original source and proof strategy. Deligne–Mumford [5, Lemmas (1.3)–(1.4), Proposition (1.5), pp. 79–82, and Theorem (5.2), pp. 104–105] analyze deformations of nodal curves and obtain the unpointed smoothness statement. Knudsen [6, Theorem 2.7, p. 179] gives the pointed theorem. Alper [4, Propositions 6.3.26 and 6.3.28, pp. 232–235; Theorem 6.4.16, p. 241] computes the pointed deformation spaces and their dimension in detail. We explain the count on the normalization and state the deformation theorem used to pass from that count to smoothness.

Our guided proof. Smoothness of a Deligne–Mumford stack over $\mathbb{C}$ can be tested on an étale atlas. Its dimension is the dimension of that atlas locally, because an étale map has relative dimension zero. To prove smoothness we use the infinitesimal lifting criterion: a small deformation over an Artinian base must lift when the base is enlarged by a square-zero ideal. A deformation of a pointed curve means a family together with an identification of its special fiber with that fixed pointed curve.

Our *external nodal deformation theorem* has two parts. First, deformations of a proper nodal curve with distinct smooth marked points have no obstruction to such infinitesimal lifting. Second, the vector space of first-order deformations fits in

$$0 \longrightarrow \bigoplus_v H^1(C_v, T_{C_v}(-B_v)) \longrightarrow \mathrm{Def}(C, p_1, \dots, p_n) \longrightarrow \bigoplus_{q \text{ a node}} T_q \longrightarrow 0, \quad (1.10.1)$$

where each T_q is a one-dimensional smoothing space. Choosing local coordinates at the two branches identifies it with K , with deformation equation $xy = \epsilon a$. That

identification depends on coordinates, but its dimension does not. The theorem is Alper's [4, Propositions 6.3.26 and 6.3.28, pp. 232–235]. The local calculation behind unobstructedness is that a node is a single equation $xy = 0$ whose deformation parameter can be lifted, while a curve has no degree-two coherent cohomology to obstruct the global gluing. The theorem makes that local-to-global argument precise; we are not developing a general obstruction theory in these notes.

The left side of (1.10.1) varies the smooth normalization components, with every marked point and every branch of a node remembered. Such variations keep the nodes unsmoothed. The right side allows one smoothing parameter for each node. Surjectivity says that these local smoothing parameters can be realized by a global first-order deformation.

Let V be the number of normalization components and let δ be the number of nodes. Riemann–Roch and the vanishing of $H^0(C_v, T_{C_v}(-B_v))$ proved above give

$$h^1(C_v, T_{C_v}(-B_v)) = 3h_v - 3 + m_v.$$

There are n marked points and two branches per node, so $\sum_v m_v = n + 2\delta$. The normalization sequence for functions gives the genus formula

$$g = \sum_v h_v + \delta - V + 1.$$

Taking dimensions in (1.10.1) now gives

$$\begin{aligned} \dim \operatorname{Def}(C, p_1, \dots, p_n) &= \sum_v (3h_v - 3 + m_v) + \delta \\ &= 3 \sum_v h_v - 3V + n + 3\delta \\ &= 3g - 3 + n. \end{aligned}$$

There are no infinitesimal automorphisms, by Proposition 1.2, so this is the tangent dimension on an étale chart, without any extra infinitesimal symmetry directions to remove. Unobstructedness and the infinitesimal lifting criterion show that the charts are smooth; the constant tangent dimension then proves the dimension assertion.

Example 1.8 (Checking the dimension at a nodal genus-two curve). For $C = E_1 \cup_q E_2$, each normalization component is elliptic with one special point. It contributes $3 \cdot 1 - 3 + 1 = 1$ parameter. The node contributes one more. The total is $1 + 1 + 1 = 3$, just as at a smooth genus-two curve. The locus where this particular node remains unsmoothed has only the first two parameters. A singular member of the family does not force the moduli stack itself to be singular.

Exercise 1.8. Glue two copies of $\mathbb{P}^1$ along three distinct pairs of points, with no markings. Check stability and genus. Use (1.10.1) to count the dimension of its deformation space. Which parameters come from the components, and which come from the nodes?

1.10.6 What depends on the hypotheses

All four propositions hold for every stable pair g, n in the setup of this chapter. The historical theorems actually prove the corresponding proper smooth Deligne–Mumford statements over $\mathrm{Spec} \mathbb{Z}$; we used their base change to $\mathbb{C}$, with the characteristic-zero expositions indicated.

If $2g - 2 + n \leq 0$, the stable pointed moduli problem as defined here has no geometric objects. One can still speak of its empty stack, but not of a nonempty smooth stack of dimension $3g - 3 + n$. If one instead drops stability while keeping those curves, the conclusions change: an unmarked $\mathbb{P}^1$ has the positive-dimensional group PGL_2 of automorphisms, and an unmarked elliptic curve has translations. Thus the argument for the Deligne–Mumford property would fail.

Finally, smoothness belongs to the *stack*; it is not an automatic assertion about its coarse moduli space. Finite group quotients of smooth charts can be singular. We have not needed the existence or projectivity of a coarse moduli space for either the stack proof or the properties proved here.

Chapter 2

$\overline{\mathcal{M}}_{g,n}(X, \beta)$ is a stack

We now keep the target variety fixed and allow a pointed curve to map to it. The new datum is an actual morphism $f : C \rightarrow X$, not merely the image curve $f(C)$. Two maps with the same image can be different objects: a double cover of a line and an isomorphism onto that line have different degrees. The arrows must remember the maps as well as their sources.

Throughout the stack proof, X is a projective scheme of finite type over $\mathbb{C}$, and A is a chosen very ample line bundle on X . Smoothness of X is not needed for this proof. Bases are arbitrary $\mathbb{C}$ -schemes, and covers are jointly surjective families of étale morphisms. For the general geometric properties below we impose the familiar, sufficient hypothesis that X is a smooth projective variety over $\mathbb{C}$. The structural curve morphism $C \rightarrow S$ is required to be of finite presentation.

2.1 The data of a stable map

Definition 2.1 (The prescribed numerical class). Start with integral curves on X and their integral linear combinations, called 1-cycles. We want to record how such a cycle meets line bundles. Two 1-cycles are numerically equivalent when they have the same degree against every line bundle on X . Denote the resulting group of integral numerical classes by $N_1(X)_{\mathbb{Z}}$. Fix a class β in this group, and put $d = \beta \cdot A$.

For a map from a proper curve $f : C \rightarrow X$, the condition “class β ” means

$$\deg_C f^*L = \beta \cdot L \quad \text{for every line bundle } L \text{ on } X.$$

For a family we impose this equality on every geometric fiber, using the pullbacks of line bundles from X . Equivalently, we fix the functional $L \mapsto \deg f^*L$ on $\text{Pic}(X)$.

This convention makes sense over every $\mathbb{C}$ -scheme without assigning a topological fundamental class to its total space. It is the degree-functional convention of Behrend–Manin, §2, following Definition 2.1, p. 12 of the arXiv version [9]. Some references instead prescribe an integral singular homology class in $H_2(X(\mathbb{C}), \mathbb{Z})$. That can contain more information: we are explicitly using the numerical version here. For $X = \mathbb{P}^2$, the class is simply the integer degree d , so the running example has exactly the usual meaning.

Degrees of an invertible sheaf in a proper flat family of curves are locally constant. One way to compute them is

$$\deg L_s = \chi(C_s, L_s) - \chi(C_s, \mathcal{O}_{C_s}).$$

Constancy of Euler characteristics in proper flat families explains both compatibility with base change and why a class condition survives descent. Later, on a finite type parameter scheme, it also identifies the fixed-class locus as a union of connected components. In that argument finitely many numerical generators suffice; alternatively, each of the finitely many connected components has a single degree functional.

Definition 2.2 (A family of stable maps). Fix integers $g, n \geq 0$. The starting data over a scheme S are

$$\xi = (\pi : C \longrightarrow S, \sigma_1, \dots, \sigma_n, f : C \longrightarrow X).$$

They form a family of stable maps of genus g , with n markings and class β , if the following conditions hold.

- (a) The morphism π is proper, flat, and of finite presentation. Its geometric fibers are connected, reduced, one-dimensional nodal curves of arithmetic genus g .
- (b) The $\sigma_a : S \rightarrow C$ are pairwise disjoint sections through the smooth locus of π .
- (c) The map f is a $\mathbb{C}$ -morphism, and its restriction to every geometric fiber has numerical class β .
- (d) On each geometric fiber, consider the normalization $\tilde{E}$ of an irreducible component E that f contracts to a point. If $\tilde{E}$ has genus 0, it must have at least three special points. If it has genus 1, it must have at least one special point. A special point is a preimage of a node or a marked point; the two branches of a self-node count separately.

There is no extra numerical restriction on a component of genus at least 2.

Here the source is allowed to have a rational component with fewer than three special points, provided the map is nonconstant on that component. Thus the source with its original markings need not be a stable pointed curve. We are directly defining the sources that occur in stable maps, without making their moduli problem a separate construction. The definitions are supported by Behrend–Manin, Definitions 3.2 and 3.4, pp. 13–15, and Fulton–Pandharipande, §1.1, pp. 10–11 [9, 8].

Over a field of characteristic zero this stability condition is equivalent to finiteness of the automorphisms preserving the markings and f . On a contracted component it is exactly pointed-curve stability. On a noncontracted component, an automorphism preserving f induces an automorphism of a finite extension of function fields, so there are only finitely many possibilities. There are also only finitely many permutations of components and branches. Conversely, a contracted rational component with at most two special points, or a contracted smooth elliptic component with no special point, has a positive-dimensional group of automorphisms preserving the map. This explains geometrically what the condition prevents.

Unlike Chapter 1, we do not require $2g - 2 + n > 0$ for all maps. When $\beta = 0$, every component is contracted, so the definition does reduce to stable pointed curves and requires the stable range for any nonempty moduli problem. When $\beta \neq 0$, $g = 0$, $n = 0$ is allowed. If no stable map has the specified class, the fibered category is empty over nonempty bases; this is permitted.

Example 2.1 (The first line map). Take $g = n = 0$, $X = \mathbb{P}^2$, and β the class of a line. Over $\text{Spec } \mathbb{C}$ an object is

$$C = \mathbb{P}^1, \quad f([u : v]) = [u : v : 0].$$

The source has no markings and is not a stable unpointed curve. Nevertheless the map is stable: its only component is not contracted. If $\alpha : C \rightarrow C$ satisfies $f\alpha = f$, then α is the identity because f is a closed immersion. The extra datum f has removed the automorphisms that made the unpointed source unstable.

2.1.1 The line bundle that will make descent work

For a nodal family the relative dualizing sheaf $\omega_{C/S}$ is an invertible sheaf, compatible with base change. The marked sections define relative effective Cartier divisors. These facts let us construct a line bundle directly from the moduli data:

$$\mathcal{L}_\xi = \omega_{C/S}(\sigma_1 + \cdots + \sigma_n) \otimes f^* A^{\otimes 3}. \quad (2.1.1)$$

“Directly” matters: no arbitrary choice of a polarization on C has been added to the object. An isomorphism of stable maps automatically identifies these line bundles. For the dualizing-sheaf input see the technical theorem used in Chapter 1, or Tag 0E6R; Fulton–Pandharipande state this ampleness criterion in §1.1, p. 11 [8].

Lemma 2.1 (Positivity and stable-map stability). *For a family satisfying the first three conditions of Definition 2.2, the fourth condition is equivalent to relative ampleness of $\mathcal{L}_\xi$.*

Proof. We have a line bundle and must test its degree on the components of a geometric fiber. If $\tilde{E}$ has genus h , has r special points, and the degree of f^*A on E is d_E , then

$$\deg_E \mathcal{L}_\xi = 2h - 2 + r + 3d_E.$$

This uses the normalization formula for the dualizing sheaf: each branch of a node contributes one to its degree. If f contracts E , then $d_E = 0$, and strict positivity is precisely the pointed-curve stability inequality. If f does not contract E , very ampleness of A gives $d_E \geq 1$, so even the smallest possible value is $-2 + 3 = 1$.

A line bundle on a proper reduced curve is ample exactly when its degree on every irreducible component is positive. We use this standard curve ampleness criterion, followed by the relative ampleness criterion for a proper finitely presented morphism: ampleness on the fiber extends to relative ampleness near that point of the base. These are the same ampleness inputs as in Chapter 1. They prove the forward implication. Conversely, relative ampleness gives positive degree on each contracted component, which forces the required special points. $\square$

For the map in Example 2.1, $\mathcal{L}_\xi = \mathcal{O}_{\mathbb{P}^1}(-2) \otimes \mathcal{O}_{\mathbb{P}^1}(3) = \mathcal{O}_{\mathbb{P}^1}(1)$. This small calculation is a preview of the general descent proof: the map to X supplies positivity on components that the markings alone do not control.

Exercise 2.1 (Check the definition). Why is a constant map from an unmarked $\mathbb{P}^1$ not stable? Why is a constant map from a smooth genus-two curve stable? Compute $\deg \mathcal{L}_\xi$ in the two cases.

2.2 Steps 1–3: build the ordinary category

2.2.1 Step 1: its objects

Let $\mathcal{K}$ initially denote an ordinary category that we are about to construct. An object is a scheme S together with a family ξ of Definition 2.2. We can write it as $(S, C, \pi, \sigma_\bullet, f)$. The scheme S is part of the object. A single geometric stable map and a one-parameter family are therefore different kinds of objects of the same total category.

Example 2.2 (A genuine family of line maps). Let $B = (\mathbb{P}^2)^\vee$, the projective plane whose points represent linear equations in the coordinates of $\mathbb{P}^2$. Inside $\mathbb{P}^2 \times B$ take the incidence variety

$$\mathcal{I} = \{([x_0 : x_1 : x_2], [a_0 : a_1 : a_2]) : a_0x_0 + a_1x_1 + a_2x_2 = 0\}.$$

Projection $\pi : \mathcal{I} \rightarrow B$ is a $\mathbb{P}^1$ -bundle; the other projection $F : \mathcal{I} \rightarrow \mathbb{P}^2$ restricts on a fiber to the inclusion of its line. Hence $(\mathcal{I} \rightarrow B, F)$ is an object of $\mathcal{K}$. It need not be a product family: the family is specified by its total space and morphisms, not by a list of unrelated fibers.

2.2.2 Step 2: its arrows

Suppose our data are two objects

$$\xi = (C \xrightarrow{\pi} S, \sigma_\bullet, f), \quad \eta = (D \xrightarrow{\rho} T, \tau_\bullet, h).$$

We want an arrow from ξ to η to say that ξ is the family obtained from η by changing the base. Thus an arrow is a pair (u, a) of scheme morphisms

$$u : S \longrightarrow T, \quad a : C \longrightarrow D$$

such that

$$\begin{array}{ccc} C & \xrightarrow{a} & D \\ \pi \downarrow & & \downarrow \rho \\ S & \xrightarrow{u} & T \end{array} \quad \begin{array}{c} \nearrow h \\ \searrow \end{array} \quad \begin{array}{c} \rho a = u\pi, \quad a\sigma_j = \tau_j u, \quad ha = f, \end{array} \quad (2.2.1)$$

and the square on the left is cartesian. The arrow to X from C is the composite $ha = f$. Equivalently, the induced map

$$\tilde{a} = (a, \pi) : C \xrightarrow{\sim} D \times_T S$$

is an isomorphism over S , taking the sections and map to X to the pulled-back ones.

This does not say that $a : C \rightarrow D$ itself is an isomorphism when $S \rightarrow T$ is not one. It says that a becomes the comparison isomorphism with the correct base-changed family. An arbitrary map of pointed source curves would not have this meaning: it could contract components or change degrees. Such maps are not arrows of this moduli category.

Example 2.3 (Reparametrization versus changing the image). Let $f([u : v]) = [u : v : 0]$ and $h([u : v]) = [v : u : 0]$. The map $a([u : v]) = [v : u]$ gives an arrow (id, a) from $(\mathbb{P}^1, f)$ to $(\mathbb{P}^1, h)$ because $ha = f$. It is the unique such arrow. If instead h parametrizes a different line, there is no arrow over $\text{Spec } \mathbb{C}$: equality $ha = f$ would force the two image lines to coincide.

For families, a map $b : S \rightarrow B$ gives a line family $\mathcal{I}_S = \mathcal{I} \times_B S$. The pair consisting of b and the projection $\mathcal{I}_S \rightarrow \mathcal{I}$ is an arrow to Example 2.2. Its base need not be a point.

2.2.3 Step 3: identities, composition, and axioms

The identity arrow of ξ is $(\text{id}_S, \text{id}_C)$. Suppose $(u, a) : \xi \rightarrow \eta$ and $(v, b) : \eta \rightarrow \zeta$, where $\zeta = (E \rightarrow U, v_\bullet, k)$. Define

$$(v, b) \circ (u, a) = (vu, ba).$$

We must check this is an allowed arrow. Commutativity over the base follows by composing the two commutative squares. The section equations give

$$ba\sigma_j = b\tau_j u = v_j vu,$$

and the target-map equations give $kba = ha = f$. Finally, pasting cartesian squares gives a cartesian square. More explicitly,

$$C \simeq D \times_T S \simeq (E \times_U T) \times_T S \simeq E \times_U S.$$

Thus composition is defined on the specified collection of arrows.

Associativity is literal associativity of the pairs of scheme morphisms: both orders of threifold composition produce the same base map and the same total-space map. The identity laws follow from the identity laws for schemes. We have therefore constructed an ordinary category, before using the words “fibered” or “stack.”

Exercise 2.2 (Check an arrow). For the two parametrizations in Example 2.3, is the identity map of $\mathbb{P}^1$ an arrow from $(\mathbb{P}^1, f)$ to $(\mathbb{P}^1, h)$? Which equation decides this? If $u : S \rightarrow T$ is not an isomorphism, must an arrow lying over u be invertible in the total category?

2.3 Steps 4–5: project to schemes and look at one fiber

2.3.1 Step 4: the projection functor

Define

$$p : \mathcal{K} \longrightarrow \text{Sch}_{\mathbb{C}}, \quad (S, C, \pi, \sigma_\bullet, f) \longmapsto S, \quad (u, a) \longmapsto u.$$

The identity and composition calculations give $p(\text{id}_\xi) = \text{id}_S$ and $p((v, b)(u, a)) = vu = p(v, b)p(u, a)$. Thus p is a functor, and $\mathcal{K}$ is a category over schemes. This is the functor called Target in your earlier terminology: it takes the target of the family morphism $\pi : C \rightarrow S$. Here the fixed target of the map $f : C \rightarrow X$ is a different scheme. The projection sends an object to S , not to X .

2.3.2 Step 5: the fiber over S

Given a fixed scheme S , the fiber category $\mathcal{K}(S)$ consists of the families over S . Its arrows are only those arrows of the total category that project to id_S . Thus an arrow in the fiber is an isomorphism $a : C \rightarrow D$ over S with

$$a\sigma_j = \tau_j, \quad ha = f.$$

Taking the fiber does not discard automorphisms and does not replace each family by its isomorphism class. In particular, even when a fiber has only one isomorphism class of objects it might still contain nonidentity arrows. Our degree-one line map happens to have trivial automorphism group; higher-degree maps need not.

Example 2.4 (An automorphism that the category remembers). The map $\mathbb{P}^1 \rightarrow \mathbb{P}^1$, $[u : v] \mapsto [u^2 : v^2]$, is a stable map with no markings. The transformation $[u : v] \mapsto [-u : v]$ is a nonidentity automorphism preserving this map. In contrast, no nonidentity reparametrization preserves an embedding onto a line in $\mathbb{P}^2$.

2.4 Steps 6–9: pullback and the cartesian property

2.4.1 Step 6: construct the pullback

We are given $\xi = (C \rightarrow S, \sigma_\bullet, f)$ and a base map $v : T \rightarrow S$. The required family over T is

$$v^*\xi = (C_T = C \times_S T \longrightarrow T, \sigma_{1,T}, \dots, \sigma_{n,T}, f_T),$$

where

$$\sigma_{j,T} = (\sigma_j v, \text{id}_T), \quad f_T = f \circ \text{pr}_C.$$

Properness, flatness, and finite presentation are preserved by base change. The pulled-back sections remain disjoint and smooth. A geometric fiber is a field extension of a geometric fiber of $C \rightarrow S$, so its genus, nodes, component degrees, and stability inequalities are preserved. Consequently this really is an object of $\mathcal{K}(T)$.

The square and target-map compatibility give an arrow

$$(v, \text{pr}_C) : v^*\xi \longrightarrow \xi.$$

The word “pullback” describes both the constructed family and this comparison arrow. It does not denote an inverse image of isomorphism classes.

Example 2.5 (Restricting the incidence family). Pulling $\mathcal{I} \rightarrow B$ back along the point $[0 : 0 : 1] \in B$ gives the line $x_2 = 0$ and its inclusion into $\mathbb{P}^2$. Pulling back along

$$\mathbb{A}^1 \longrightarrow B, \quad t \longmapsto [-t : 0 : 1]$$

gives the family $x_2 = tx_0$. Its fibers are distinct lines; its total space and map to $\mathbb{P}^2$ keep track of how they vary.

2.4.2 Step 7: verify the universal property

A cartesian arrow is characterized by a factorization problem. Given an arrow $q : \xi_T \rightarrow \xi$ lying over $v : T \rightarrow S$, it must have the following property: for any object ζ over U , any map $w : U \rightarrow T$, and any arrow $r : \zeta \rightarrow \xi$ over vw , there exists a unique arrow $\tilde{r} : \zeta \rightarrow \xi_T$ over w such that $q\tilde{r} = r$. The base factorization $U \xrightarrow{w} T \xrightarrow{v} S$ is part of the given data.

Apply this definition to $\xi_T = v^*\xi$. Write $\zeta = (E \xrightarrow{\lambda} U, \epsilon_\bullet, e)$ and write $b : E \rightarrow C$ for the total-space map of r . We must construct a map to $C \times_S T$. Its two required components are forced:

$$\tilde{b} = (b, w\lambda) : E \longrightarrow C \times_S T.$$

They fit together because $\pi b = vw\lambda$. The ordinary universal property of the scheme fiber product gives existence and uniqueness of $\tilde{b}$.

There are still three moduli checks. First,

$$\tilde{b}\epsilon_j = (b\epsilon_j, w) = (\sigma_j vw, w) = \sigma_{j,T}w.$$

Second, $f_T \tilde{b} = fb = e$, so the arrow respects the fixed target. Third, because r was an arrow, $E \simeq C \times_S U$; using w we obtain

$$(C \times_S T) \times_T U \simeq C \times_S U \simeq E.$$

Thus its square over w is cartesian. The pair $(w, \tilde{b})$ is the required arrow, and no other one can have the same composite with q . This completes the categorical universal-property check, including the additional map to X .

Exercise 2.3 (Check the extra equation). In this factorization problem, why is it insufficient merely to construct a scheme morphism $\tilde{b} : E \rightarrow C \times_S T$? Name the conditions still required for it to define an arrow of stable-map families, and verify the equation involving X .

2.4.3 Step 8: the fibers are groupoids

An arrow $a : \xi \rightarrow \eta$ over id_S is an isomorphism of source schemes over S . Its inverse preserves the markings, since $a\sigma_j = \tau_j$ implies $a^{-1}\tau_j = \sigma_j$. It preserves the target maps, since $ha = f$ implies $fa^{-1} = h$. Therefore a^{-1} is also an arrow in the fiber. Every arrow in $\mathcal{K}(S)$ is invertible: each fiber is a groupoid.

This is a statement about arrows over a fixed base. For example, the arrow from one line fiber to the entire incidence family lies over $\text{Spec } \mathbb{C} \rightarrow B$ and is not invertible in the total category.

2.4.4 Step 9: the fibered-category conclusion

For every base map and object above its target we have constructed a cartesian lift, and the fibers are groupoids. Thus $p : \mathcal{K} \rightarrow \text{Sch}_{\mathbb{C}}$ is a category fibered in groupoids. In this particular category every arrow is cartesian, since by definition its source is isomorphic to the base change of its target; transporting the universal property across that isomorphism proves the assertion.

Iterated pullbacks need not be literally equal as schemes. There is instead a canonical isomorphism $(C \times_S T) \times_T U \simeq C \times_S U$, compatible with sections and maps to X . This is why a category fibered in groupoids is the appropriate language for change of base. It records all these identifications without demanding artificial equalities of chosen fiber products.

2.5 Step 10: isomorphisms form a sheaf

Fantechi's Definition 4.3 requires two further statements, after the category fibered in groupoids has been constructed. The first is Definition 4.2: compatible local isomorphisms between two already existing global objects must come from a unique global isomorphism. The second, Definition 4.1, concerns constructing a global object from local objects and transition isomorphisms. We now check the first condition for maps, spelling out the target compatibility [1, §4.2–4.3, Definitions 4.1–4.3].

2.5.1 What the sheaf assigns to a test scheme

Given two families over S ,

$$\xi = (C \rightarrow S, \sigma_\bullet, f), \quad \eta = (D \rightarrow S, \tau_\bullet, h),$$

define a presheaf on schemes over S by

$$\underline{\text{Isom}}_S(\xi, \eta)(T \rightarrow S) = \{a : C_T \xrightarrow{\sim} D_T : a\sigma_{j,T} = \tau_{j,T}, h_T a = f_T\}.$$

Restriction is base change. The sheaf condition says that an étale cover of any such T detects equality of these isomorphisms and permits gluing compatible ones. It is enough to carry out the argument over a scheme called S , since the same argument can then be applied to every $T \rightarrow S$.

2.5.2 The given local arrows

Let $\{S_i \rightarrow S\}$ be an étale cover and write $S_{ij} = S_i \times_S S_j$. Suppose we have isomorphisms of stable maps

$$a_i : C_i = C \times_S S_i \xrightarrow{\sim} D_i = D \times_S S_i$$

such that, after the canonical pullback identifications,

$$a_i|_{S_{ij}} = a_j|_{S_{ij}}.$$

Each a_i respects the markings and satisfies $h_i a_i = f_i$. We must construct an isomorphism $a : C \rightarrow D$ over S with these restrictions, not simply a bijection between geometric fibers.

2.5.3 Glue the underlying morphism

The maps $C_i \rightarrow C$ form an étale cover, obtained by base change from $S_i \rightarrow S$. Compose a_i with $D_i \rightarrow D$. These are compatible morphisms from the covering schemes C_i to the scheme D . The representable functor $\text{Hom}(-, D)$ is an étale sheaf: morphisms of schemes satisfy faithfully flat descent. Hence there is a unique scheme morphism $a : C \rightarrow D$ restricting to them. This is the morphism-gluing theorem of Chapter 1; a precise technical reference is Tag 023Q.

The equalities making a a morphism over S hold after the same cover, so they hold globally by the uniqueness part of morphism descent. This theorem also glues the a_i^{-1} to a morphism $b : D \rightarrow C$. The composites ba and ab restrict to identity morphisms on the cover. Uniqueness gives $ba = \text{id}_C$ and $ab = \text{id}_D$. Thus a is an isomorphism.

2.5.4 Check the markings and the target

For each j , the two morphisms $a\sigma_j, \tau_j : S \rightarrow D$ agree on every S_i , hence are equal. The new issue in the maps problem is the equality of the two morphisms

$$ha, f : C \longrightarrow X.$$

Their restrictions to every C_i agree because a_i is a map isomorphism. Morphism descent, now with target X , implies $ha = f$. Therefore the glued scheme isomorphism is an isomorphism of stable maps. Any other candidate has the same underlying morphism after the cover, so must equal a . We have proved exactly existence and uniqueness required by the sheaf condition.

Example 2.6 (Compatible isomorphisms of line families). Suppose two line families over S have locally chosen parametrizations $p_i : \mathbb{P}_{S_i}^1 \rightarrow C_i$ and $q_i : \mathbb{P}_{S_i}^1 \rightarrow D_i$. A local isomorphism is expressed in these coordinates by $q_i m_i p_i^{-1}$ for a projective linear change of parameter m_i . For it to be an arrow it must satisfy

$$h_i q_i m_i = f_i p_i.$$

On overlaps the coordinate expressions may change, but the resulting morphisms $C_{ij} \rightarrow D_{ij}$ must agree. The sheaf proof glues these actual morphisms. It does not require the matrices for the m_i to have identical entries in different trivializations. For a line embedded in $\mathbb{P}^2$ the compatible map is forced by the embeddings whenever it exists.

Exercise 2.4 (Why the target equation is essential). Suppose local isomorphisms of the source curves agree on overlaps but are not required to satisfy $h_i a_i = f_i$. What does morphism descent still give? Why does that result not prove the sheaf condition for isomorphisms of stable maps?

2.6 Step 11: descent data for objects are effective

2.6.1 The data and the goal

The previous proof began with two global families. We now start with no global family at all. Fantechi's descent datum supplies an étale cover $\{S_i \rightarrow S\}$, a family

$$\xi_i = (C_i \rightarrow S_i, \sigma_{1,i}, \dots, \sigma_{n,i}, f_i : C_i \rightarrow X)$$

on every member of the cover, and transition isomorphisms

$$\alpha_{ij} : C_i \times_{S_i} S_{ij} \xrightarrow{\sim} C_j \times_{S_j} S_{ij}$$

of stable-map families over S_{ij} . In particular,

$$\alpha_{ij}\sigma_{a,i} = \sigma_{a,j}, \quad f_j\alpha_{ij} = f_i. \quad (2.6.1)$$

On S_{ijk} we require the cocycle equation

$$\alpha_{ik} = \alpha_{jk} \circ \alpha_{ij}. \quad (2.6.2)$$

Restrictions to overlaps in these equations are understood. The equation says that transporting a point, a section, or any other piece of structure from chart i to chart k directly agrees with transporting it through chart j . The isomorphism and cocycle requirements include identity on the diagonal and inverse transition maps; equivalently these follow by restricting the usual cocycle equations appropriately.

Effectivity asks us to construct a family $\xi = (C \rightarrow S, \sigma_\bullet, f)$ and isomorphisms

$$\theta_i : C \times_S S_i \xrightarrow{\sim} C_i$$

such that

$$\alpha_{ij} = \theta_j \theta_i^{-1} \quad \text{over } S_{ij}.$$

We need a scheme C , a flat proper family, markings, and a map f ; after those constructions we must verify the prescribed class and stability. Merely asserting that the geometric fibers are familiar curves would not perform any of these tasks.

2.6.2 Why the underlying schemes can be descended

An arbitrary descent datum for schemes need not be effective as a scheme. Thus we use the same specific theorem as in Chapter 1: descent is effective for proper finitely presented schemes carrying relatively ample invertible sheaves with compatible descent isomorphisms. This is polarized effective descent. Vistoli, §4.3.3, Theorem 4.38 and the discussion on pp. 96–103, develops the underlying faithfully flat descent argument; the relatively ample form is also SGA 1, Exposé VIII, Proposition 7.8, as recorded in the source map [2, 3].

We now verify its hypotheses, rather than assuming that the curves already descend. On each C_i form

$$\mathcal{L}_i = \omega_{C_i/S_i}(\sigma_{1,i} + \cdots + \sigma_{n,i}) \otimes f_i^* A^{\otimes 3}.$$

By Lemma 2.1, $\mathcal{L}_i$ is relatively ample. An isomorphism α_{ij} canonically identifies relative dualizing sheaves and marked divisors. The equality $f_j\alpha_{ij} = f_i$ identifies $\alpha_{ij}^* f_j^* A$ with $f_i^* A$. These identifications combine to give

$$\alpha_{ij}^* \mathcal{L}_j \simeq \mathcal{L}_i.$$

They satisfy the cocycle condition: every factor is defined functorially from the maps and markings, and the α_{ij} satisfy (2.6.2). We have genuine compatible line-bundle isomorphisms, not only equal numerical degrees.

All local curves are proper, flat, and finitely presented over their bases. The cover is étale, hence flat and locally of finite presentation. To apply a theorem stated for a faithfully flat quasi-compact morphism, work over an affine open of S . Refine the cover by affine opens in the S_i . Their images are open and cover the quasi-compact base, so finitely many suffice. The disjoint union of these finitely many affine schemes gives the required faithfully flat quasi-compact cover. After performing descent there, glue over the affine opens of S using uniqueness of compatible descent isomorphisms. Thus arbitrary bases and arbitrary étale covering families cause no hidden quasi-compactness assumption.

The theorem produces a scheme $C \rightarrow S$ with the desired θ_i , and a descended relatively ample line bundle. Flatness, properness, and finite presentation descend under faithfully flat base change. The relative family therefore has precisely these properties. This use of an ample bundle is the substantive geometric difference between a purely formal cocycle and an effective scheme descent datum.

For intuition behind the theorem, over a quasi-compact piece one takes sufficiently large compatible powers of the ample bundles, descends the modules of sections and their multiplication, and reconstructs the curve by relative Proj of the resulting graded algebra. The sheaf-descent and relative-Proj theorems are substantial inputs. We use the polarized descent theorem as a black box; the preceding argument verifies why this moduli datum lies in its domain of application.

2.6.3 Descend the marked sections

Use θ_i^{-1} to regard each $\sigma_{a,i}$ as a section of $C \times_S S_i \rightarrow S_i$. The first equality in (2.6.1) says that these local sections agree on overlaps. Morphism descent now gives a unique map $\sigma_a : S \rightarrow C$. Its composite with $C \rightarrow S$ is the identity because this is true after pulling back to each S_i . Hence it is a section. Membership in the smooth locus and pairwise disjointness can be checked after a surjective étale base change, so the sections have the required properties.

2.6.4 Descend the maps to X

Through θ_i , the morphisms f_i become morphisms from the cover $C \times_S S_i$ of C to X . The second equality in (2.6.1) says exactly that these morphisms agree on their double

overlaps. Morphism descent gives a unique

$$f : C \longrightarrow X.$$

This step does not require X to be smooth. The representable functor of a scheme is a sheaf, so morphisms to that scheme glue. Projectivity of X entered earlier, to construct the compatible relatively ample $\mathcal{L}_i$ that guaranteed scheme effectivity.

Notice the order: the local maps first supplied line bundles with descent identifications; after descending the polarized sources we glued the maps themselves. We have not assumed a global f in order to construct the scheme on which f will be defined.

2.6.5 Check the geometric conditions and the class

Given a geometric point $\bar{s} \rightarrow S$, choose a lift to a member of the étale cover, after an algebraically closed field extension if necessary. Pulling back θ_i identifies the resulting geometric fiber of $C \rightarrow S$ with that of $C_i \rightarrow S_i$, together with its markings and map. Hence it is connected, reduced, nodal, and of genus g . Each contracted component satisfies the required special-point inequality. These properties are preserved and detected by algebraically closed field extension.

For every line bundle L on X , the same comparison gives

$$\deg_{C_{\bar{s}}} f_{\bar{s}}^* L = \deg_{(C_i)_{\bar{s}_i}} f_{i,\bar{s}_i}^* L = \beta \cdot L.$$

Thus the class is β . All defining conditions have now been checked, so the constructed tuple is an object of $\mathcal{K}(S)$. The maps θ_i are isomorphisms of stable maps and reproduce the prescribed transitions. This is effectivity in precisely Fantechi's sense.

Example 2.7 (A descent datum of line maps). Take the incidence family over $B = (\mathbb{P}^2)^\vee$. Over the standard affine opens B_i , its source $\mathbb{P}^1$ -bundle can be trivialized. In a chosen trivialization we have maps

$$F_i : \mathbb{P}_{B_i}^1 \longrightarrow \mathbb{P}^2.$$

On B_{ij} , let α_{ij} be the change from the i -parametrization to the j -parametrization. Then

$$F_j \alpha_{ij} = F_i, \quad \alpha_{jk} \alpha_{ij} = \alpha_{ik}.$$

These are the actual transition and cocycle equations for stable-map objects. Descent recovers the incidence family and its projection to $\mathbb{P}^2$. We already know the answer in this example, which makes it a useful test of the definition. The proof above, unlike the example, applies when a global answer was not known in advance.

One can see a nontrivial change of parameter even on a constant line family. Cover $\mathbb{A}^1$ by $U = D(t)$ and $V = D(t - 1)$. On U use $F_U([u : v]) = [u : v : 0]$; on V use $F_V([u : v]) = [v : u : 0]$. On the overlap the transition is $[u : v] \mapsto [v : u]$. The equality $F_V \alpha_{UV} = F_U$ makes this a descent datum; forgetting that equality would lose the information needed to glue the map to $\mathbb{P}^2$.

Exercise 2.5 (Locate the hypotheses). In the effective-descent proof, where did projectivity of X enter? Where did the cocycle condition enter? Where did we use compatibility with f_i ? Explain why the stability of the underlying pointed curves from Chapter 1 cannot simply be assumed here.

2.7 Step 12: the stack conclusion

Theorem 2.1. *Let X be a projective scheme of finite type over $\mathbb{C}$, let $g, n \geq 0$, and prescribe a numerical curve class β as in Definition 2.1. The category of stable-map families and cartesian, marking-preserving, map-preserving arrows is a stack on $\text{Sch}_{\mathbb{C}}$ for the étale topology. We denote it by*

$$\overline{\mathcal{M}}_{g,n}(X, \beta).$$

Proof. Steps 1–9 construct a category fibered in groupoids. Step 10 proves that isomorphisms form a sheaf. Step 11 proves that every descent datum is effective. These are the two extra conditions in Fantechi’s Definition 4.3. In particular the notation now refers to a stack, rather than to a hoped-for moduli space or a set of isomorphism classes. $\square$

2.8 Properties

The stack proof is complete before the next propositions begin. Algebraicity asks for a scheme atlas; the Deligne–Mumford property asks for an étale atlas; properness controls extension across missing points of bases. None of these follows just from the sheaf and object-descent conditions. We use the conventions for algebraic, Artin, and Deligne–Mumford stacks explained in Chapter 1. In particular a Deligne–Mumford stack is an algebraic (Artin) stack with the stronger étale-atlas property.

Proposition 2.1 (Algebraicity and finite type). *For a smooth projective variety $X/\mathbb{C}$, fixed integers $g, n \geq 0$, and fixed numerical class β , the stack $\overline{\mathcal{M}}_{g,n}(X, \beta)$ is algebraic and of finite type over $\mathbb{C}$. No stable-range inequality is required beyond the stability condition defining its objects. Empty cases are included.*

Original source and proof strategy. Behrend–Manin, Theorem 3.14, p. 25, with the construction in §4, p. 28, gives the algebraic finite type stack statement for smooth projective targets and the degree-functional notion of class [9]. This is an early research construction. Their proof uses additional pointed curves. Fulton–Pandharipande, §2.3, pp. 13–14, gives a more direct bounded Hilbert-scheme construction that we use to explain an atlas [8]. Their primary output is a coarse moduli space, and §1.2 explicitly leaves the stack interpretation to the reader. Consequently the quotient-stack identification below is our guided reconstruction from their parameter scheme, not a theorem we attribute verbatim to that exposition. We make no claim here to establish absolute historical priority.

Our guided proof. Embed $X \hookrightarrow \mathbb{P}^r$ using A . A stable map to X becomes a stable map of degree $d = \beta \cdot A$ to $\mathbb{P}^r$, with the same stability condition. We want one finite type scheme that parametrizes every such map after a choice of coordinates on its source. If the moduli problem is empty, the claim is immediate, so we may assume that maps exist.

The boundedness input in Fulton–Pandharipande, §2.3, says that an integer $q > 0$ can be chosen depending only on g, n, r, d so that, for every geometric stable map, $\mathcal{L}_\xi^{\otimes q}$ is very ample and has vanishing H^1 . This is a substantial uniform theorem: ordinary ampleness for each individual curve is not enough to give one parameter space. Its numerical output is

$$e = q(2g - 2 + n + 3d), \quad N + 1 = e + 1 - g.$$

Riemann–Roch gives $h^0(C, \mathcal{L}_\xi^{\otimes q}) = N + 1$. Cohomology and base change show that in families $E = \pi_* \mathcal{L}_\xi^{\otimes q}$ is a vector bundle of this rank and its formation commutes with base change. These are black-box inputs in this atlas construction.

Locally choose projective coordinates on E . Evaluation embeds C into $\mathbb{P}_S^N$. Taking the graph of its map to $X \subset \mathbb{P}^r$ gives

$$C \hookrightarrow \mathbb{P}_S^N \times \mathbb{P}_S^r.$$

With respect to $\mathcal{O}(1, 1)$, its Hilbert polynomial is

$$P(m) = (e + d)m + 1 - g.$$

Fixing this polynomial places all these graphs on one projective Hilbert scheme. The marked points are added using the incidence subscheme of the Hilbert scheme times $(\mathbb{P}^N \times \mathbb{P}^r)^n$.

We then impose the conditions defining the family: nodal connected fibers; distinct smooth markings; projection to $\mathbb{P}^N$ an embedding given by the complete linear series;

and the required relation between $\mathcal{O}(1)$ and $\mathcal{L}_\xi^{\otimes q}$. Fulton–Pandharipande, §2.2, Proposition 1, and §2.3, construct the relevant locally closed parameter scheme J . We take that parameter-scheme construction as a black box, including the assertion about its scheme structure. The equality of line bundles is imposed up to a line bundle pulled back from the base, which is exactly what projective coordinates allow.

Requiring that the graph lie in $\mathbb{P}^N \times X$ is a closed Hilbert-scheme condition. Finally retain the components of numerical class β . Fiberwise line-bundle degrees are locally constant, and this parameter scheme has only finitely many connected components. Hence this is an open and closed locus of a finite type scheme. Continue to denote the resulting scheme by J .

The group $G = \mathrm{PGL}_{N+1}$ changes the projective coordinates on the source embedding. The original map to X and its markings do not change as abstract moduli data. For an arbitrary family ξ/S , the scheme of projective frames of E is a G -torsor $P \rightarrow S$, and the embedded family supplies a G -equivariant map $P \rightarrow J$. Conversely, a G -torsor and such an equivariant map give a family on the torsor whose transition isomorphisms satisfy descent. The stack proof already completed descends it to S . Both constructions preserve arrows and are inverse up to canonical isomorphism. Thus

$$\overline{\mathcal{M}}_{g,n}(X, \beta) \simeq [J/G].$$

The brackets record a quotient *stack*. They retain the stabilizers of points of J , which are exactly automorphisms of the corresponding stable maps.

For a smooth algebraic group acting on a scheme, $J \rightarrow [J/G]$ is a smooth surjective atlas: after pulling back along the object $(P \rightarrow S, P \rightarrow J)$ it becomes the torsor $P \rightarrow S$. This is Fantechi’s quotient-stack construction and Theorem 5.3 [1]. Since J and G are finite type over $\mathbb{C}$, the quotient stack is of finite type. The atlas has supplied the extra geometric structure that the word “stack” alone did not supply.

Proposition 2.2 (The Deligne–Mumford property and separatedness). *Under the hypotheses of Proposition 2.1, the stack is Deligne–Mumford and separated. More precisely, its diagonal is representable, finite, and unramified.*

Original source and proof strategy. Behrend–Manin, Proposition 4.1 and Lemma 4.2, p. 26, establish this diagonal statement by using additional intersections with general hyperplanes to compare map isomorphisms with isomorphisms of stable pointed curves [9]. The argument below explains that method with explicit bookkeeping for permutations of the auxiliary markings. The criterion that an algebraic stack with unramified diagonal is Deligne–Mumford is an external general theorem about algebraic stacks, as in the references for Chapter 1.

Our guided proof. Start with two stable-map families over S . Pulling the diagonal back by this pair gives their isomorphism sheaf. We must show this sheaf is represented by a finite unramified S -scheme; the sheaf condition alone did not say that.

Work étale locally near a geometric point of S and use $X \hookrightarrow \mathbb{P}^r$. Choose sufficiently many general hyperplanes, at least three, avoiding the images of special points and of contracted components. Since the characteristic is zero, the maps on noncontracted components are generically separable; the hyperplanes can meet these maps transversely. Their inverse images are disjoint smooth points on the source fibers. After shrinking and an étale base change, these finite étale divisors split into labeled sections. Add them to the original markings of both source families.

Each formerly unstable noncontracted rational component now has at least three extra points, and each noncontracted elliptic component has at least one. Contracted components were stable already. Thus the enlarged pointed sources are stable curves, so Chapter 1 applies to their isomorphism schemes.

A map isomorphism preserves the inverse image of every chosen hyperplane as a divisor. It may permute the labels within that divisor. We must therefore allow finitely many permutations, rather than insisting that all map isomorphisms preserve one arbitrary labeling. For each such permutation, consider the isomorphism scheme of the two enlarged stable pointed curves with the corresponding reordered labels. Each is finite and unramified by the diagonal theorem for stable curves. Their finite disjoint union parametrizes all the possible underlying isomorphisms that could preserve the maps.

Within that union impose $h\alpha = f$. It is a closed condition. Indeed the two morphisms to projective X define two graphs in the relevant relative Hilbert scheme, and equality is the pullback of its closed diagonal; equivalently one uses the separatedness of the relative Hom-scheme. The representability of Hilbert/Hom functors in this projective setting is an external theorem already used for the atlas. The resulting closed subscheme is precisely the map-isomorphism sheaf. A closed subscheme of a finite unramified S -scheme is again finite and unramified. These properties and representability descend from the étale neighborhood.

If the degree is zero, no auxiliary points are needed: the sources are themselves stable pointed curves, and the same closed compatibility condition cuts out the required isomorphisms. In every case we have proved the diagonal statement. Finiteness implies properness of the diagonal, which is the separatedness condition. Algebraicity plus an unramified diagonal gives the Deligne–Mumford property.

The characteristic-zero hypothesis is meaningful here. For example, in characteristic p the map $\mathbb{P}^1 \rightarrow \mathbb{P}^1$, $[u : v] \mapsto [u^p : v^p]$, has nontrivial infinitesimal automorphisms:

over $k[\epsilon]/(\epsilon^2)$, replacing u by $u + \epsilon v$ leaves u^p unchanged. Thus the unrestricted positive-characteristic analogue need not have unramified diagonal. Behrend–Manin’s theorem includes a degree bound relative to the characteristic to exclude this issue. Our choice of $\mathbb{C}$ avoids it.

Proposition 2.3 (Properness). *For X smooth projective over $\mathbb{C}$ and g, n, β fixed as above, $\overline{\mathcal{M}}_{g,n}(X, \beta)$ is proper over $\mathbb{C}$.*

Original source and proof strategy. Behrend–Manin, Theorem 3.14 and Corollary 4.8, pp. 25 and 28, give properness of the stack; the latter refers to Pandharipande’s preliminary notes for the extension step [9]. We did not locate and verify that preliminary proof independently. Fulton–Pandharipande’s later accessible exposition, §4.2, Propositions 5 and 6, pp. 19–21, proves the relevant uniqueness and existence statements for families [8]. Its strategy is semistable reduction together with extension of the map to a projective target. We use the stable-map extension theorem as a black box, rather than developing the separate constructions involved in proving it.

Our guided proof. Properness asks whether a family over the generic point of a discrete valuation ring can acquire a limit. The stack version allows a finite extension of the fraction field and a dominating valuation ring; this allowance is needed for stable reduction. Existence of such a limit, uniqueness of isomorphisms between limits, and finite type together give the valuative criterion for a separated Deligne–Mumford stack.

The needed black box is the following form of stable-map reduction: a stable map over the fraction field of a discrete valuation ring, with projective target, extends after a finite extension to a stable-map family over a dominating discrete valuation ring. Isomorphisms of generic-fiber stable maps extend uniquely between two such families. For the general statement we invoke Behrend–Manin’s theorem; Fulton–Pandharipande give the geometric-curve version and its proof strategy. The reduction from that version to the usual valuative criterion in this finite type characteristic-zero setting is part of the general properness machinery, not a consequence of descent alone.

Here is why the hypotheses match. Embed X in $\mathbb{P}^r$ and apply extension for maps to $\mathbb{P}^r$. If a resulting family has generic map in X , its whole map lies in X . To see the scheme-theoretic assertion, let $\mathcal{I}_X$ be the ideal of X and consider its image in the structure sheaf of the flat source over the valuation ring. That image vanishes on the generic fiber. A flat module over a discrete valuation ring has no nonzero element killed by a power of its uniformizer, so the image is zero. This is stronger than just saying that the special-fiber points lie in X .

The class remains β because degrees of pullbacks of line bundles are constant in the resulting proper flat family. The isomorphism uniqueness statement is also reflected in the finite diagonal established above. Finally, Proposition 2.1 provides finite type. Applying the valuative criterion gives properness.

Projectivity, or an appropriate replacement theorem for a proper target, cannot simply be omitted from this proof. For a nonproper target properness already fails among constant maps: $\overline{\mathcal{M}}_{0,3}(\mathbb{A}^1, 0) \simeq \mathbb{A}^1$. A constant map here means a three-pointed stable rational curve together with its chosen image point, which can escape to infinity. The example isolates why existence of a limit is additional geometry beyond the stack conditions.

Proposition 2.4 (A smooth stable-map stack, with an explicit hypothesis). *For $X = \mathbb{P}_{\mathbb{C}}^2$, $g = n = 0$, and β the class of a line, there is an equivalence of stacks*

$$\overline{\mathcal{M}}_{0,0}(\mathbb{P}^2, 1) \simeq (\mathbb{P}^2)^{\vee}.$$

In particular this stable-map stack is a smooth proper scheme of dimension 2.

Original source and proof strategy. The description of degree-one maps by lines is elementary; our proof below verifies it on families, which is stronger than identifying geometric points. Fulton–Pandharipande, §1.2, Theorem 2 and the subsequent stack comment, p. 12, and §5.2, pp. 27–30, include smoothness in genus zero for projective-space targets [8]. We use no general smoothness theorem to prove this particular proposition and make no originality claim for the elementary identification.

Our guided proof. First examine a geometric object. Its genus-zero nodal source has a tree as dual graph, and every component has normalization $\mathbb{P}^1$. Degree is nonnegative on each component, and total degree 1 means that exactly one component is noncontracted. There can be no other components: a finite tree with more than one vertex has at least two leaves, at most one of which can be the noncontracted vertex. A contracted leaf would have one node and no markings, contradicting stability. Thus the source is a smooth $\mathbb{P}^1$, and the degree-one map embeds it as a line.

Now let the base be an arbitrary scheme S . Every geometric fiber of $\pi : C \rightarrow S$ is a smooth $\mathbb{P}^1$, so the proper flat finitely presented family is smooth. Set $L = f^* \mathcal{O}_{\mathbb{P}^2}(1)$. On every fiber it is $\mathcal{O}_{\mathbb{P}^1}(1)$, with $h^0 = 2$ and $h^1 = 0$. Cohomology and base change give a rank-two vector bundle $E = \pi_* L$, compatible with base change. The evaluation $\pi^* E \rightarrow L$ is surjective by fiberwise generation and Nakayama’s lemma, so it defines a map $e : C \rightarrow \mathbb{P}_S(E)$, using the quotient convention for the projective bundle. We check that this is an isomorphism on the whole family, including over a nonreduced base. The map is proper and finitely presented; each of its geometric fibers over S

is the isomorphism $\mathbb{P}^1 \rightarrow \mathbb{P}^1$ defined by the complete degree-one linear series. Thus it has finite fibers and is finite (Tag 02LS, Lemma 37.44.1). The fiberwise flatness criterion makes e flat: its source is flat and locally finitely presented over S , its target is locally of finite type over S , and its fiber maps are flat (Tag 039E, Lemma 37.16.4). A finite flat finitely presented morphism is finite locally free (Tag 02K9, Lemma 29.49.2). Its rank here is one, so the unit map $\mathcal{O}_{\mathbb{P}(E)} \rightarrow e_*\mathcal{O}_C$ is a map of line bundles that is an isomorphism on each fiber, hence an isomorphism by Nakayama. Therefore e is an isomorphism. These are scheme-theoretic fiber criteria, not an inference from a bijection of points alone.

Pulling back the three coordinate sections of $\mathcal{O}_{\mathbb{P}^2}(1)$ gives a surjective map

$$\mathcal{O}_S^{\oplus 3} \longrightarrow E.$$

It is surjective because it is surjective on every geometric fiber, followed by Nakayama's lemma locally on S . A rank-two quotient of this trivial rank-three bundle is exactly a morphism $S \rightarrow \mathrm{Gr}(2, 3) = (\mathbb{P}^2)^\vee$. The original curve and map are the pullback of the universal incidence line under this morphism.

Conversely, a map $S \rightarrow (\mathbb{P}^2)^\vee$ pulls back the incidence family to a stable-map family. These constructions commute with base change and are inverse up to canonical isomorphism. An automorphism over S compatible with the inclusion in $\mathbb{P}_S^2$ is the identity, because a closed immersion is a monomorphism. Thus the equivalence records arrows as well as objects; it is an equivalence of stacks, not just a bijection of closed points. Smoothness, properness, and dimension 2 now follow from the corresponding properties of a projective plane.

Stable-map stacks are not smooth in general, even for smooth projective targets. The smoothness assertion just proved has the explicit degree-one, genus-zero hypotheses in its statement. An accessible account of the failure in genus one is Battistella–Carocci–Manolache, §8.3, equation (8.2), p. 29 [10]: near maps with a contracted elliptic component and one rational tail, local charts have equations $\xi y_1 = \cdots = \xi y_r = 0$ in a smooth ambient chart. Where $\xi = y_1 = \cdots = y_r = 0$, this local model has intersecting components and is singular. We cite that local-equation calculation as an external result; we do not derive it or use it in the stack proof. Their §8.2, Example 8.4, p. 27, also illustrates different-dimensional components for genus-one degree-three maps to $\mathbb{P}^2$. The dimension 2 above is an ordinary actual dimension, obtained from an explicit scheme description.

Exercise 2.6 (Why lines have no hidden boundary). Explain why a genus-zero degree-one stable map to $\mathbb{P}^2$ with no markings cannot have any contracted components. Why would the argument need to be reconsidered if markings were allowed?

Chapter 3

Connection to the cuspidal thesis problem

The preceding chapters prove statements about the two specified nodal moduli problems. They also identify the work needed for a new moduli problem of maps whose source curves may have ordinary cusps. They do not construct such a moduli stack by replacing one word in a definition.

3.1 What transfers formally, and what needs geometry

An ordinary cusp over an algebraically closed field of characteristic zero has completed local ring

$$k[[x, y]]/(y^2 - x^3) = k[[t^2, t^3]].$$

Unlike a node, it has one branch on normalization. The normalization parameterization is $x = t^2, y = t^3$. This local model specifies the allowed singularity; it does not specify a stability condition for an entire pointed curve or map.

To begin the thesis construction, first write down the objects: proper flat finitely presented source families, allowed geometric fibers, distinct smooth labelled sections, a map to the fixed target, the meaning of β , and an explicit stability condition. We must know whether an elliptic subcurve, a contracted cuspidal component, or a rational component with too few attachments is allowed. None of those decisions is determined by the word “cusp.”

Once these data are chosen, the earlier work separates into the following tasks.

- (a) *Formal category theory.* Use pairs (u, h) with cartesian source squares, preservation of markings, and $f = gh$. Identities and composition have the same explicit

formulas as before. The projection retains the parameter scheme. Its fibers are groupoids because the arrows over an identity are required to be isomorphisms. These checks need no new theorem about cusps.

- (b) *Base change of the proposed objects.* The cartesian-arrow calculation transfers once pulling back an object really stays inside the proposed class. Properness, flatness, and finite presentation already behave correctly. One must verify the geometric singularity and stability conditions over arbitrary geometric fibers and arbitrary base changes. A condition stated only for closed complex points is not yet enough to define our category over all schemes.
- (c) *The sheaf condition for isomorphisms.* The underlying morphism theorem is unchanged. Glue h_i , glue h_i^{-1} , and check the equations for the base, sections, and target map after the cover. Once objects are well defined, this part does not depend on a node having two branches. It also does not establish that the isomorphism sheaf is representable or unramified.
- (d) *Effective descent.* Here there is a real geometric input to supply. Nodes and ordinary cusps are hypersurface singularities, hence Gorenstein. For a proper flat finitely presented family with these geometric fibers, the relative-dualizing theorem used earlier therefore still supplies an invertible $\omega_{C/S}$ and its base-change identifications. Smooth markings still give relative Cartier divisors. The candidate bundle

$$\omega_{C/S}(\sum \sigma_a) \otimes f^* A^{\otimes q}$$

is consequently functorial for any fixed integer q . What needs a proof for the *chosen* stability condition is its relative ampleness, or some other effective-descent mechanism. The nodal formula counting two node branches cannot simply be copied to a cusp. If a suitable functorial ample bundle is found, the same polarized descent theorem constructs the total space; the maps and sections then descend as before. The permitted singularities and the proposed stability condition must be recoverable after the chosen covers as well.

- (e) *Algebraicity and the Deligne–Mumford property.* These are further theorems. One needs a bounded embedding result and a representable parameter locus for the permitted cuspidal families and maps (or another algebraicity argument). One must prove representability of the diagonal and control infinitesimal automorphisms. Finiteness of automorphisms, and the role of characteristic zero, must be justified for the new stability condition. Canonical ampleness alone no longer gives the nodal argument; see the example below.

- (f) *Finite type, separatedness, and properness.* Finite type requires a uniform bound for the chosen discrete data. Separatedness requires uniqueness of an extension of an isomorphism. Properness requires existence of admissible limits after suitable finite base changes, with the desired stability condition, and compatibility with the target map. Ordinary stable reduction proves these statements for the nodal compactification; it is not a proof of them for a different cuspidal compactification. One must prove or locate the corresponding theorems for the exact new moduli problem.

In particular, the geometry needed for object descent is different from the geometry needed for a finite-type atlas or for a unique limit. Keeping these obligations separate tells us which claim has actually been established at any point in the thesis.

3.2 A small warning supplied by an actual cusp

Example 3.1 (Ample pointed dualizing bundle with infinitely many automorphisms). Consider the projective cuspidal cubic

$$C = \{y^2z = x^3\} \subset \mathbb{P}_{\mathbb{C}}^2, \quad p = [0 : 1 : 0].$$

The point p is smooth and the cusp is $[0 : 0 : 1]$. The normalization is $\mathbb{P}^1$. At the cusp, the quotient $\mathbb{C}[[t]]/\mathbb{C}[[t^2, t^3]]$ has dimension one, with basis the class of t . The normalization sequence therefore gives $\chi(\mathcal{O}_C) = \chi(\mathcal{O}_{\mathbb{P}^1}) - 1 = 0$, so the arithmetic genus is one. As a hypersurface, C is Gorenstein, and its dualizing sheaf is invertible. The general Gorenstein-curve formula $\deg \omega_C = 2p_a(C) - 2$, Stacks Project Lemma 53.8.3, Tag 0C19, gives degree zero. Thus $\omega_C(p)$ has degree one and is ample, by the componentwise ampleness criterion Tag 0B5Y. This calculation does not presume that the curve is nodal.

Nevertheless, for every $\lambda \in \mathbb{C}^*$,

$$[x : y : z] \longmapsto [\lambda^2 x : \lambda^3 y : z]$$

preserves C and fixes p . These are infinitely many distinct automorphisms, forming a positive-dimensional group. They also give nonzero infinitesimal automorphisms by allowing $\lambda = 1 + \epsilon$ with $\epsilon^2 = 0$.

This object is not asserted to belong to any proposed pseudostable moduli problem. It shows concretely why the nodal equivalence between the stability inequalities, canonical ampleness, and finite automorphisms needs reconsideration when cusps are admitted. For a constant map $C \rightarrow X$, these automorphisms still preserve the

map. For the inclusion $C \hookrightarrow \mathbb{P}^2$, an automorphism preserving that inclusion must be the identity. The target map can change the automorphism calculation dramatically.

Exercise 3.1. Suppose you have proved a functorial ample-bundle theorem for your proposed cuspidal maps and that all defining conditions are local for étale covers. Which of Fantechi's two conditions can you now prove? Which further assertions are still required before calling the result a proper Deligne–Mumford stack? Use the cuspidal cubic to explain why the second question is not answered merely by ampleness.

The next thesis step is therefore to specify the exact cuspidal stability condition and test these obligations against it. These notes establish neither the intended pseudostable-map stack nor a virtual class for it. The later theories named in the project brief are deliberately left for subsequent notes.

Appendix A

Solutions to the checks

A.1 Stable pointed curves

Solution to Exercise 1.1

The normalization has two points above the self-node and one marked point, so $r = 3$. For a rational component, $2h - 2 + r = -2 + 3 = 1 > 0$. Counting the node only once would give $r = 2$, which would incorrectly declare the component unstable. The local dualizing contribution comes from both branches.

Solution to Exercise 1.2

The base map is $i : \operatorname{Spec} \mathbb{C} \rightarrow \mathbb{A}^1$ at 0 and the total-space map is $C_0 \rightarrow C_0 \times \mathbb{A}^1$ at 0. The square is a fiber-product square, so this is a moduli arrow. If it had an inverse in the total category, applying p would give an inverse to i , which does not exist. A fiber category over S contains only arrows over id_S . The arrow just constructed lies over i , not over an identity, so it is not a counterexample to the groupoid assertion.

Solution to Exercise 1.3

Apply the universal property of $\xi_T \rightarrow \xi$ to $\xi'_T \rightarrow \xi$ with base factorization id_T to obtain $b : \xi'_T \rightarrow \xi_T$. Interchanging the roles gives $c : \xi_T \rightarrow \xi'_T$. Both bc and the identity are factorizations of the same arrow through $\xi_T \rightarrow \xi$, so uniqueness gives $bc = \operatorname{id}$; similarly $cb = \operatorname{id}$. The commuting condition determines this isomorphism. Without that condition, the identity and a hyperelliptic involution can be two different isomorphisms between the same genus-two objects.

Solution to Exercise 1.4

Morphism descent produces a map, whose invertibility still needs verification. Glue the compatible local inverse maps and check the two composite maps against the identities after the covers of the two total spaces. Uniqueness of morphism gluing then makes them identities globally. For each label r , compare the maps $a\sigma_r$ and τ_r from the base to the target total space. Their local equality gives global equality. Invertibility alone would not say that labels are preserved.

Solution to Exercise 1.5

The canonical base-change isomorphisms for ω and preservation of every marking divisor give $a_{ij}^* L_j \simeq L_i$. Denote this isomorphism by λ_{ij} . Functoriality carries the equation $a_{ik} = a_{jk} a_{ij}$ to

$$\lambda_{ik} = \lambda_{ij} \circ a_{ij}^* \lambda_{jk}.$$

Thus the bundle transitions have the required cocycle. Unrelated ample bundles need not even pull back to isomorphic bundles under the specified a_{ij} ; and arbitrary chosen isomorphisms, when available, need not satisfy a cocycle. The canonical bundle supplies both existence and coherence without adding a new choice to the moduli object.

Solution to Exercise 1.6

An overlap point is a pair of square roots u_1, u_2 of the same nonzero t . Either $u_2 = u_1$ or $u_2 = -u_1$. Only the first comparison is on the diagonal. Two sign changes compose to the identity since $\iota^2 = \text{id}$; in coordinates $(u_2/u_3)(u_1/u_2) = u_1/u_3 = 1$. From $y_i = z/u_i$ one gets $y_j = (u_i/u_j)y_i$, exactly the prescribed transition map. Checking only the diagonal of this overlap would lose all the information about the involution.

Solution to Exercise 1.7

For a rational normalization component, stability gives $m_v \geq 3$, so $\deg T_{\mathbb{P}^1}(-B_v) = 2 - m_v < 0$. For an elliptic component, $m_v \geq 1$ and the degree is $-m_v < 0$. For $h_v \geq 2$, the degree $2 - 2h_v - m_v$ is already negative even if $m_v = 0$. An unmarked elliptic curve instead has trivial tangent bundle and a nonzero translation vector field. A finite group may have several elements: the hyperelliptic involution of a smooth genus-two curve gives such an example. Triviality means that only the identity remains, which is a stronger condition than the one needed for a Deligne–Mumford stack.

Solution to Exercise 1.8

There are two rational normalization components, each with three special points, so both are stable. The genus is $0 + 0 + 3 - 2 + 1 = 2$. Each component contributes $3 \cdot 0 - 3 + 3 = 0$ parameters: a three-pointed rational curve has no moduli. Each of the three nodes contributes one smoothing parameter. The deformation space therefore has dimension 3, in agreement with $3g - 3 = 3$.

A.2 Stable maps**Solutions for Chapter 2****Solution to Exercise 2.1**

A contracted rational component with no special points violates the requirement of three special points. Its canonical polarization has degree -2 . A constant map from a smooth genus-two curve is stable because its only component has genus at least two; its polarization has degree $2g - 2 = 2$. The pullback of A under a constant map has degree zero in both cases.

Solution to Exercise 2.2

The identity of $\mathbb{P}^1$ does not give this arrow: it would require $h \circ \text{id} = f$, whereas the two displayed parametrizations are different maps. The coordinate swap is the required arrow. An arrow over a noninvertible base map cannot be invertible in the total category, since any inverse would project to an inverse of the base map. Its comparison with the base-changed target is nevertheless an isomorphism.

Solution to Exercise 2.3

The total-space map must lie over w , preserve the marked sections, preserve the map to X , and induce an isomorphism with the base-changed target family. For the last target equation, $f_T = f \text{pr}_C$ and $\text{pr}_C \tilde{b} = b$, so $f_T \tilde{b} = fb = e$. The product description $E \simeq C \times_S U \simeq (C \times_S T) \times_T U$ proves the cartesian-square condition.

Solution to Exercise 2.4

The compatible source isomorphisms glue to a unique isomorphism of the source schemes over S , and to a pointed-curve isomorphism if the markings were respected

locally. Without $h_i a_i = f_i$ there is no reason for $ha = f$. The result would therefore be an arrow in a different moduli problem. For example, the identity of a source $\mathbb{P}^1$ does not identify two different parametrizations of a line as stable maps unless the maps themselves agree.

Solution to Exercise 2.5

Projectivity supplies the very ample A and hence the relatively ample $\omega(\sum \sigma_i) \otimes f^* A^3$. The cocycle equation makes the curve and line-bundle identifications consistent on triple overlaps, so that the polarized descent theorem applies. Compatibility with f_i identifies the $f_i^* A^3$ factors and later lets the maps to X themselves glue. Underlying pointed curves need not be stable: an unmarked degree-one $\mathbb{P}^1$ is already a counterexample. It is stable as a map because its nonconstant map contributes positivity and removes source automorphisms preserving the map.

Solution to Exercise 2.6

The dual graph is a tree with exactly one vertex carrying positive degree. If it has at least two vertices, at least one leaf has degree zero. That contracted leaf has a single node and no markings, contradicting the three-special-point requirement for a contracted rational component. With markings, a contracted leaf might have one node and two markings, and would then be stable. Thus the absence of markings is an essential hypothesis in the argument.

A.3 The cuspidal roadmap

Solution to Exercise 3.1

The isomorphism-sheaf argument glues underlying morphisms, their inverses, and the compatibility equations. Effective object descent then uses the functorial ample bundle to descend the total spaces, followed by the markings and the map, and the assumed locality to recover all defining conditions. Along with the category and base-change checks, this proves the stack axioms.

It does not construct an algebraic atlas, prove boundedness, represent the diagonal, exclude infinitesimal automorphisms, or prove existence and uniqueness of limits. Those are needed for the proposed proper Deligne–Mumford conclusion. The marked cuspidal cubic has an ample $\omega_C(p)$ but a positive-dimensional automorphism group,

demonstrating that even the automorphism part does not follow from ampleness in the cuspidal setting.

Appendix B

Source map and scope of the reconstruction

This map was assembled before the chapters were drafted. References were checked in the passages listed below, including the local Fantechi transcription and the local Behrend–Manin text. Page numbers refer to the linked version unless explicitly called journal pages. This is a map of what each source actually supplies, not a claim that one reference proves every step of these notes.

Categories, pullbacks, and the two stack conditions.

Fantechi, *Stacks for Everybody* [1], Sections 3.2, 3.5, 4.1–4.3 and 6.3: Definition 3.1 (source PDF p. 5), Definition 4.1 (p. 6), Definition 4.2 (p. 6), Definition 4.3 (p. 7). The local file is `../.. /Papers/Stacks/StacksForEverybody.tex`, relative to this directory. Its labels are `def:3.1`, `def:4.1`, `def:4.2`, and `def:4.3`. This is a later pedagogical exposition using bundles; our curves and maps are applications, not examples whose full proofs appear in that article.

Gluing morphisms, and hence isomorphisms.

Vistoli, *Notes on Grothendieck topologies, fibered categories and descent theory* [2], Section 4.3, Proposition 4.31, pp. 91–92, and the representable-sheaf theorem used there; Stacks Project [11], Tag 023Q, Lemma 35.13.7. The latter checks the exact statement: scheme-valued representable functors are fpqc sheaves. We apply it on a cover of the *source total space*, glue the inverse separately, and check markings and target maps by equality after a cover. No representability theorem for the isomorphism functor is needed to prove this stack condition.

Effective descent of the total spaces.

SGA 1 [3], Exposé VIII, Section 7, Proposition 7.8, is the scheme-with-ample-invertible-sheaf descent theorem. Vistoli, Section 4.3.3, Theorem 4.38 and its proof, pp. 96–103, gives a detailed later explanation using a functorial relatively ample invertible sheaf. His Example 4.39 treats smooth curves, not our entire pointed nodal/map problem. Our reconstruction supplies the extra checks: the functorial bundles $\omega_{C/S}(\sum \sigma_a)$ and $\omega_{C/S}(\sum \sigma_a) \otimes f^* A^{\otimes 3}$, their cocycles, and descent of the sections and of f . Vistoli, Proposition 1.15, pp. 11–12, records descent of the required morphism properties. Stacks Project Tag 0E6R, Lemma 109.19.3, checks the invertibility and arbitrary-base-change property of the relative dualizing sheaf for Gorenstein curve families.

Stable curves, embeddings, and geometric properties.

Deligne–Mumford [5], Section 1, Definition (1.1), Theorem (1.2) and its corollary, and Lemma (1.4), journal pp. 76–81, give the unpointed stability/embedding/automorphism ingredients; Section 5, Proposition (5.1) and Theorem (5.2), pp. 104–105, give the unpointed stack and its properness and smoothness. Knudsen [6], Theorem 2.7, journal p. 179, is the pointed proper smooth Deligne–Mumford theorem. These are original research sources whose indicated statements were checked. Alper [4], in the dated 5 January 2026 version, Propositions 6.3.23–6.3.28, pp. 231–235, Theorems 6.4.6 and 6.4.16, pp. 238–241, and Section 6.5, pp. 242–252, supplies the later detailed exposition. Our atlas proof applies his embedding-lifting argument directly to stable pointed families. Edidin [7], Section 3, offers a supplementary unpointed Hilbert-scheme exposition. The guided proofs explicitly retain Hilbert representability, bounded embeddings, deformation theory of nodal curves, and stable reduction as external theorems. They are not deductions from the two stack axioms alone.

Stable maps: definitions and bounded parameter spaces.

Fulton–Pandharipande [8], Section 1.1, pp. 10–11, gives families, isomorphisms, and the ample stability bundle; Section 2.3, pp. 13–14, gives bounded embeddings and the Hilbert incidence construction. Section 5.1, Lemma 8, pp. 26–27, handles factorization through the target. This is an exposition and primarily a construction of *coarse spaces*; Section 1.2 explicitly sets aside the stack viewpoint. Identifying the corresponding quotient *stack*, and verifying Fantechi’s axioms, are our guided reconstruction with the descent references above. Behrend–Manin [9], Definition 2.1, p. 12, supports the degree-functional convention for β ; Definitions 3.2 and 3.13, pp. 13–14 and 25, give stable-map and stack notation.

Definition 3.13 asserts the stack property without supplying the step-by-step descent proof requested here.

Stable maps: Deligne–Mumford, finite type, and properness.

Behrend–Manin, Theorem 3.14, p. 25, Proposition 4.1 and Lemma 4.2, pp. 26–27, and Corollary 4.8, p. 28, support these properties for the stated projective target and characteristic assumptions. This is an early research treatment; no absolute priority claim is needed here. Its properness argument invokes earlier preliminary notes. Fulton–Pandharipande, Section 4.2, Propositions 5 and 6, pp. 19–21, supplies readable proofs about extending isomorphisms and families, respectively. The latter uses semistable reduction as an external theorem. Our properness argument passes from those family statements to the stack valuative criterion. Smoothness is established in the explicit degree-one plane example by a direct identification with the dual projective plane; no general smoothness assertion is made.

B.1 How the search was evaluated

The local search included all of `DoctoralNotebook/Papers/`: in particular the Alper PDF, Fantechi transcription, Behrend–Manin, Behrend’s Gromov–Witten paper, the Faber–Pandharipande text, and the later virtual-class and pseudostability material. The last groups were not used as substitutes for foundations or brought into the present scope. The earlier *BG* and Riemann–Roch summaries supplied the memoir/header conventions, checkpoints, and exercise format.

The public search used the suggested authors as leads and also found Vistoli, Edidin, Schmitt, and student seminar notes. It distinguished foundational construction from later work that uses these stacks. Supplementary sources and their limitations are recorded in Section B.2. No advisor–student relationship is asserted. When a proof below invokes a substantial theorem, it is labelled as an external input rather than concealed in a phrase such as “the objects glue.”

B.2 Supplementary reading and proof boundaries

The following evaluations explain why some promising references were useful as reading companions but not chosen as the proof of a stack axiom.

Vakil and the Pixton course lead.

Vakil [12, Sections 2.4–2.7, pp. 9–11] gives geometric motivation for stable curves

and their compactification. Pixton's Math 697 course reference page also directs readers to a Vakil article for an informal introduction. This is useful orientation, not a proof of object descent. Vakil's later 2022 course notes, Section 45.6, p. 88, give another stable-map algebraicity route using a mapping stack, but leave part of its representability discussion as an exercise. The direct family and Hilbert methods fit the present scope better.

Zvonkine.

The stable-curve discussion in [13, Section 1.4, pp. 14–18] is particularly good for examples of compactification and special-point stability. Its account is intentionally informal about foundations, so it does not replace the scheme descent proof.

Student expositions.

Belfiori [14, Section 1.1, pp. 2–5] explicitly describes cartesian family arrows and the role of a canonical polarization in descent, citing Alper and Vistoli. Its theorem about the stable compactification's geometric properties is stated rather than proved. Monavari [15, Sections 1.2–1.3, pp. 11–16] gives additional practice with fibered categories and Deligne–Mumford stacks; it does not prove both of our complete constructions. These were read as supplementary expositions, not as original sources for the classical theorems.

Niven Achenjang's *Introduction to Stacks and Moduli Notes* for Alper's Winter 2021 course, pp. 19–21 and Sections 5.2 and 5.6, pp. 24 and 28–29, gives a conversational companion to descent and Hilbert presentations. We used the checked Alper text for the formal results. Cavalieri's *Moduli Spaces of Pointed Rational Curves*, Fields graduate summer school, July 2016, Section 1.2, pp. 10–11, discusses families and their equivalence; its genus-zero focus does not supply the two general stack proofs.

Other leads and inaccessible material.

The search also followed the suggested Cavalieri, Caporaso, Farkas, Graber, Pandharipande, Pixton, and Ross leads. Several results concerned intersection theory, particular components, or later applications of moduli stacks, which does not establish the foundational gluing claim. For example, Farkas's *Regular components of moduli spaces of stable maps* directs readers to Fulton–Pandharipande for construction. A search-indexed Imperial lecture-note file advertised a stable-map algebraicity section but the actual file could not be retrieved; no claim here relies on that unseen section. The UCLA student stack seminar supplied

another reading list, but the handwritten PDFs could not be inspected through the available web reader and were not used as evidence.

B.2.1 What has and has not been reconstructed

The two stack proofs are complete relative to their stated scheme-theoretic inputs: morphism descent, duality and base change, ampleness criteria, and polarized effective descent. The hypotheses of those inputs are checked at their uses. For the later properties, Hilbert representability and bounded embeddings, relevant infinitesimal lifting results, criteria for Deligne–Mumford stacks and properness, and the stable reduction/extension theorems are substantial black boxes. The guided proofs explain how they imply the propositions; they do not replace the full proofs of those external theorems.

We did not find one accessible pedagogical source spelling out all the requested pointed-curve and stable-map descent checks in Fantechi’s notation. The present chapters supply that reconstruction from the verified general descent theorems. Behrend–Manin’s reference to an earlier preliminary properness proof was not independently traced to an accessible text; Fulton–Pandharipande supplies a verifiable later family-level argument. No historical-priority assertion is made for that unverified preliminary source.

There is no unresolved classical stack claim in the stated setup once the declared external results are accepted. There is also no claim to have constructed the intended cuspidal moduli problem: its defining stability condition, descent polarization, boundedness, diagonal, and limit theorems are the separate research tasks in Chapter 3.

References

- [1] Barbara Fantechi, *Stacks for Everybody*, in *European Congress of Mathematics*, Barcelona 2000, Vol. I, Progress in Mathematics 201, Birkhäuser, 2001, pp. 349–359. DOI and published chapter. Local transcription: `../.. /Papers/Stacks/StacksForEverybody.tex`. The source-PDF page markers in that file are used for its internal pagination. Sections 3.2, 3.5, 4.1–4.3, 5, and 6.3 are used here.
- [2] Angelo Vistoli, *Notes on Grothendieck topologies, fibered categories and descent theory*, author version of 2 October 2008. Author PDF; arXiv record. The notes form the descent contribution to *Fundamental Algebraic Geometry: Grothendieck’s FGA Explained*, AMS, 2005. Numbering and pages in these notes refer to the 2008 author PDF: Propositions 1.15, 4.31, 4.37; Theorems 4.23 and 4.38.
- [3] Alexander Grothendieck et al., *Revêtements étales et groupe fondamental (SGA 1)*, Documents Mathématiques 3, Société Mathématique de France, 2003, recomposed and annotated edition of Lecture Notes in Mathematics 224 (1971). arXiv edition, PDF. Exposé VIII, Proposition 7.8 is on printed p. 174 of the recomposed edition (PDF p. 190; original margin pp. 224–225).
- [4] Jarod Alper, *Stacks and Moduli*, version of 5 January 2026. Dated author PDF. Local copy: `../.. /Papers/Stacks/JarodAlperModuliNotes.pdf`. The dated version is important: its stable-curve material is Chapter 6, and theorem numbering differs from earlier versions. Principal passages: Sections 6.3–6.5, pp. 225–252; Theorems 2.1.2, 4.2.1, 4.7.1, and 4.8.7; Appendix C.
- [5] Pierre Deligne and David Mumford, *The irreducibility of the space of curves of given genus*, Publications Mathématiques de l’IHÉS 36 (1969), 75–109. Numdam record; original PDF. Sections 1 and 5; especially Theorem (1.2), Lemma (1.4), Proposition (5.1), and Theorem (5.2). Pages in the text are journal pages.

- [6] Finn F. Knudsen, *The projectivity of the moduli space of stable curves, II: The stacks $M_{g,n}$* , *Mathematica Scandinavica* **52** (1983), 161–199. DOI and journal article. Theorem 2.7, journal p. 179, is the pointed proper smooth stack theorem. Its historical “algebraic stack” terminology is the Deligne–Mumford terminology used here.
- [7] Dan Edidin, *Notes on the construction of the moduli space of curves*, in *Recent Progress in Intersection Theory*, Trends in Mathematics, Birkhäuser, 2000, pp. 85–113. arXiv record and text. Section 3 of the preprint, Theorems 3.2–3.4, pp. 15–17, is a later exposition of the unpointed construction and reduction arguments; it is supplementary rather than the main proof here.
- [8] William Fulton and Rahul Pandharipande, *Notes on stable maps and quantum cohomology*, in *Algebraic Geometry—Santa Cruz 1995*, Proceedings of Symposia in Pure Mathematics 62, Part 2, AMS, 1997, pp. 45–96. arXiv record; PDF. Citations use the preprint’s printed pagination: Sections 1–2, pp. 10–14; Section 4.2, Propositions 5–6, pp. 19–21; Section 5.1, Lemma 8, pp. 26–27; Section 5.2, pp. 27–30.
- [9] Kai Behrend and Yuri Manin, *Stacks of stable maps and Gromov–Witten invariants*, *Duke Mathematical Journal* **85** (1996), no. 1, 1–60. arXiv record; PDF. Local text: `../.. /Papers/BehrendManin/BehrendManin.tex`. Citations use arXiv pagination: Definitions 2.1, 3.2, 3.4, 3.13; Theorem 3.14; Proposition 4.1, Lemma 4.2, and Corollary 4.8, pp. 12–28.
- [10] Luca Battistella, Francesca Carocci, and Cristina Manolache, *Virtual Classes for the Working Mathematician*, *SIGMA* **16** (2020), 026, 38 pages. DOI; journal PDF. Local text: `../.. /Papers/VCWM/sigma20-026.tex`. Only the concrete singular stable-map example is used: Section 8.2, Example 8.4, p. 27; Section 8.3, equation (8.2), p. 29. The paper’s later-purpose machinery is not used in our stack proofs.
- [11] The Stacks Project Authors, *The Stacks Project*, project website. Tags are permanent identifiers and are linked at their uses. Main technical checks: Tag `023Q` (morphism descent), Tag `0E6R` (relative dualizing sheaf), Tag `0B5Y` (curve ampleness), Tag `0D2S` (relative ampleness), Tag `02L0`, Tag `02L1`, Tag `02L2` (descent of morphism properties). The project fills precise technical gaps here; Fantechi, Vistoli, Alper, and Fulton–Pandharipande provide the main pedagogical route.

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- [12] Ravi Vakil, *The moduli space of curves and Gromov–Witten theory*, in *Enumerative Invariants in Algebraic Geometry and String Theory*, Lecture Notes in Mathematics 1947, Springer, 2008, pp. 143–198. arXiv record; author preprint. Sections 2.4–2.7, preprint pp. 9–11, give the supplementary compactification discussion considered here.
 - [13] Dimitri Zvonkine, *An introduction to moduli spaces of curves and their intersection theory*, version of 2 November 2014. dated lecture-note PDF. Section 1.4, pp. 14–18, gives geometric intuition; its introductory account deliberately does not develop the stack foundations.
 - [14] Filippo Belfiori, *On the Construction of Some Stacks of Curves and Surfaces*, master’s thesis, Università di Bologna, academic year 2023–24. institutional thesis PDF. Section 1.1, Definition 1.6, Remark 1.7, Theorem 1.12, Remark 1.13, and Theorem 1.18, pp. 2–5.
 - [15] Sergej Monavari, *Local Picture of Twisted Curves with an Introduction to the Theory of Deligne–Mumford Stacks*, ALGANT master’s thesis, 2018. institutional thesis PDF. Sections 1.2–1.3, pp. 11–16, were considered for introductory fibered-category and stack definitions.